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THE DEVELOPMENT OF AN EXPERIMENTAL BENCHMARK FOR
HYBRID QUANTUM-CLASSICAL NEURAL NETWORKS USING KERNEL
METHODS

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ABSTRACT

While hybrid quantum-classical neural networks theoretically offer superior representational capacity in high-dimensional Hilbert spaces, empirical benchmarks frequently fail to isolate quantum utility from classical artifacting. Consistent with recent literature, we find that quantum models do not systematically outperform classical baselines across all regimes. To identify specific conditions for quantum advantage, this thesis introduces a strict 1-to-1 parameter-ablated benchmark across a 2×2 empirical matrix of data topology (tabular vs. geometric) and volume (sufficient vs. extreme $p \gg N$ starvation).

We identify four distinct operational boundaries: (1) a geometric advantage under data starvation, where quantum models (SplitAttentionQNN) outperform parameter-matched classical equivalents by 24.48 percentage points on topologically complex manifolds ($p = 6.5 \times 10^{-9}$, Welch t -test, $n = 20$ seeds); (2) representational parity under multi-omics feature starvation, where residual quantum architectures (ResQNet, QKAResQNet) match classical network capacity with statistically confirmed parity ($p > 0.47$) and tighter variance; (3) standard regime parity, where classical models dominate strictly due to execution time efficiency ($\times 1,900$ – $\times 54,000$ faster per epoch with no accuracy loss); and (4) a classical decoherence boundary, where all hybrid architectures fail on quantum-native entangled data ($\approx 50\%$ accuracy) due to irreversible entanglement destruction by the classical compression layer. Furthermore, SplitAttentionQNN exhibits statistically significant underperformance on genomic data (Leukemia: $p = 0.0016$; Colon: $p = 0.0098$), empirically confirming that inductive bias misalignment actively degrades performance beyond mere parity failure.

These findings establish that inductive bias alignment between the parameterized quantum circuit and the data manifold is a necessary condition for NISQ-era advantage. The experimental setup enforces a strict 1-to-1 replacement strategy where classical and hybrid model twins share identical ingestion layers, scaling mechanisms, optimizers, and total parameter counts ($\Delta \leq 28$ parameters, $< 0.1\%$). All quantum simulations incorporate the empirically derived noise profile of the 133-qubit IBM Torino processor (single-qubit error: 0.03%; two-qubit error: 0.62%; readout error: 3.03%). Results are validated across up to 20 random

seeds per experiment using Welch's t -test and Mann-Whitney U to ensure robust statistical generalization.

Keywords: Quantum Machine Learning, Hybrid Neural Networks, ResQNet, Data Starvation, Parameterized Quantum Circuits, Inductive Bias, NISQ.

INTRODUCTION

Relevance of the research topic. The rapid scaling of classical machine learning models has increasingly encountered thermal and computational bottlenecks. Concurrently, Quantum Machine Learning (QML) has emerged as a promising alternative, theoretically offering superior representational capacity through the exploitation of high-dimensional complex Hilbert spaces. In the Noisy Intermediate-Scale Quantum (NISQ) era, Parameterized Quantum Circuits (PQCs) integrated into hybrid classical-quantum architectures represent the most viable path toward achieving practical utility [1; 2].

However, the field of QML is currently facing an epistemological benchmarking crisis. Despite significant theoretical advancements, empirically quantifying the advantage of NISQ-era models remains challenging [3]. The literature frequently suffers from methodological inconsistencies, where classical and quantum models are compared using different optimization strategies, parameter scales, or data splits. Furthermore, a critical objective remains unaddressed in empirical practice: identifying the exact operational boundaries where quantum architectures provide measurable advantage.

Degree of development of the research topic. Existing hybrid frameworks typically deploy quantum circuits sequentially within classical architectures, without systematic ablation strategies [4]. The “qubit bottleneck”—the strict limitation on the number of qubits available in NISQ devices—forces information-theoretic compression that frequently destroys non-local correlations [5]. Current benchmarking methodologies exhibit selection bias by testing quantum models primarily on trivial, data-rich manifolds instead of pathological regimes where quantum advantage might manifest [6]. Recent work emphasizes the importance of inductive bias alignment and data encoding sensitivity [7; 8].

The problem to be solved. The catastrophic interpolative collapse (overfitting) of classical neural networks under extreme data starvation ($p \gg N$) remains inadequately studied in the context of hybrid quantum-classical architectures. There exists no standardized, reproducible benchmark to measure quantum resilience against this collapse. Specifically: (1) What operational boundaries define genuine quantum advantage? (2) Does inductive bias alignment between

circuit structure and data topology predict quantum utility? (3) How do NISQ hardware errors interact with gradient flow in hybrid architectures?

Table 1 — Research Gap: This Thesis vs. Prior Quantum Machine Learning Benchmarks

Dimension	Bowles et al.	Huang et al.	Caro et al.	Basilewitsch	This Work
Parameter parity (1-to-1)	No	No	No	No	Yes
Hardware noise profile	No	No	No	Partial	Yes
Extreme $p \gg N$ regime	No	Partial	No	No	Yes
Topology-aware datasets	No	No	Yes	Partial	Yes
Cross-domain evaluation	No	No	Partial	Yes	Yes

Table 1 synthesizes the research landscape: while prior works addressed individual dimensions (parameter matching, noise profiles, or topological evaluation), no prior benchmark systematically evaluated all five criteria. This thesis fills this gap through comprehensive ablation across all dimensions, providing a unified framework for evaluating quantum advantage under realistic constraints.

Goal of the research. The goal of this thesis is to develop a strict, symmetric benchmarking methodology to evaluate hybrid quantum-classical networks against mathematically equivalent classical baselines, and to identify the specific conditions under which quantum models provide measurable advantage.

Tasks of the research. To achieve this goal, the following tasks were formulated and solved:

1. Develop a software pipeline using PyTorch and Qiskit with strict 1-to-1 parameter ablation methodology.
2. Design and implement hybrid architectures (ResQNet, SplitAttention-QNN) and classical baselines with identical parameter counts.
3. Conduct computational experiments across a 2×2 empirical matrix of data topology and volume regimes.
4. Simulate realistic hardware noise from the 133-qubit IBM Torino processor with empirically derived error profiles.
5. Identify and characterize four distinct operational boundaries for hybrid model performance.

Object of the research. Hybrid quantum-classical neural network architectures and their computational mechanisms within machine learning frameworks.

Subject of the research. The representational capacity, topological inductive bias alignment, and noise robustness of the ResQNet architecture under extreme data starvation ($p \gg N$) and simulated NISQ hardware errors.

Scientific novelty. This thesis presents the first comprehensive, symmetrically-ablated cross-domain benchmark for hybrid QML models. Key novel contributions include: (1) the SplitAttentionQNN architecture for processing high-dimensional inputs within qubit constraints; (2) empirical characterization of inductive bias alignment as a necessary condition for quantum advantage; (3) identification of the “Classical Decoherence Boundary,” proving that classical compression irreversibly destroys quantum entanglement before the quantum circuit can process it; (4) systematic evaluation of NISQ hardware noise as a dual-function artifact that initially regularizes but eventually destroys model capacity.

Practical significance. The ResQNet and SplitAttentionQNN architectures are evaluated on real-world biomedical datasets (Leukemia, Colon Cancer genomics) and molecular cheminformatics (QSAR), demonstrating applicability to industry-relevant problems. The identified operational boundaries provide actionable guidance for practitioners choosing between classical and hybrid approaches based on data characteristics.

Methodology and research methods. This research employs strict 1-to-1 parameter ablation, Principal Component Analysis (PCA) for topological latent space visualization, and the Qiskit TorchConnector for automatic differentiation via the parameter-shift rule. All experiments enforce identical data-loading seeds, optimization strategies, and hardware noise profiles. Statistical robustness is ensured through 20-seed and multi-split evaluation protocols.

Provisions for defense. The following main provisions are submitted for defense:

1. Inductive bias alignment between the parameterized quantum circuit and the data manifold is a necessary condition for NISQ-era quantum advantage. On topologically complex manifolds (Swiss Roll), SplitAttentionQNN achieves 91.92% accuracy versus 67.43% for classical equivalents, demonstrating 24+ percentage point advantage [9].
2. The “Classical Decoherence Boundary” demonstrates that standard hybrid architectures are fundamentally incapable of processing quantum-

native entangled states due to classical compression bottlenecking [5]. All hybrid models on the NTangled dataset stalled at 50% (random guessing).

Degree of reliability and approbation of results. Reliability is ensured by the rigorous ablation methodology, use of established datasets (OpenML, UCI), empirically calibrated hardware noise from IBM Torino, and automated hyperparameter tracking. All results are reproducible via open-source PyTorch/Qiskit implementations with version-controlled code.

1 THEORETICAL FOUNDATIONS OF QUANTUM MACHINE LEARNING AND THE CURSE OF DIMENSIONALITY

1.1 The Epistemological Challenge in Quantum Machine Learning Benchmarks

The intersection of quantum information science and statistical learning theory has catalyzed the development of Quantum Machine Learning (QML). However, the field is currently grappling with a profound epistemological challenge regarding the empirical demonstration of “quantum advantage” [1]. The foundational critique frequently leveled against QML research is that classical deep learning architectures—leveraging highly optimized graphical processing units and decades of algorithmic refinement—can routinely solve standard classification tasks with near-perfect accuracy [4].

When a classical Multilayer Perceptron (MLP) or Residual Network (ResNet) achieves 100% accuracy on a dataset at a fraction of the temporal and financial cost, justifying the integration of noisy intermediate-scale quantum (NISQ) devices becomes structurally difficult [10].

This critique, however, is fundamentally an artifact of selection bias in benchmarking methodologies [3]. Evaluating a hybrid quantum-classical architecture on trivial manifolds or highly abundant datasets (where the number of samples approaches infinity) mathematically obfuscates the inherent capabilities of quantum Hilbert spaces [6]. Classical machine learning models are universal approximators that thrive in data-rich environments; they interpolate efficiently when the sample density adequately covers the high-dimensional feature space [2]. The genuine utility of quantum computation in machine learning does not reside in achieving higher raw accuracy on dense, perfectly separable classical datasets [11].

Instead, the quantum advantage manifests critically in pathological data regimes: extreme data starvation, ultra-high-dimensional feature spaces where the number of features vastly exceeds the number of available samples ($p \gg N$), and environments requiring the extraction of highly complex, non-local correlations [5].

To construct a fully convincing, unassailable experimental benchmark for hybrid quantum-classical neural networks, the evaluation framework must pivot away from simplistic accuracy metrics on abundant data. The paradigm must be redefined around the mathematical concepts of topological expressivity, sample complexity, information geometry, and the unique implicit regularization properties that quantum circuits exhibit under realistic hardware noise [12; 13].

1.2 Quantum Computing Fundamentals: From Classical Bits to Quantum Advantage

1.2.1 Classical vs. Quantum Computation: Core Distinction

The foundational distinction between classical and quantum computation lies not merely in engineering complexity, but in the mathematical structures that underpin information representation and manipulation. A classical bit is confined to a binary state: $b \in \{0, 1\}$. In contrast, a quantum bit (qubit) can exist in a superposition of both states simultaneously, existing as a continuous point on the Bloch sphere, a three-dimensional unit sphere that geometrically represents all possible single-qubit states.

While a classical bit is discrete and definite, a qubit exists in a complex superposition:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1 \quad (1)$$

The Bloch sphere representation parameterizes this state using spherical coordinates:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle \quad (2)$$

where θ (polar angle) and ϕ (azimuthal angle) uniquely specify every valid quantum state. The north pole represents $|0\rangle$ and the south pole represents $|1\rangle$; points on the equator represent equal superpositions.

The genuine power of quantum computation emerges when scaling to N qubits. The state space grows as 2^N -dimensional complex vector space, in sharp contrast to N bits which can represent only one of 2^N possible values at any given time. An N -qubit system is:

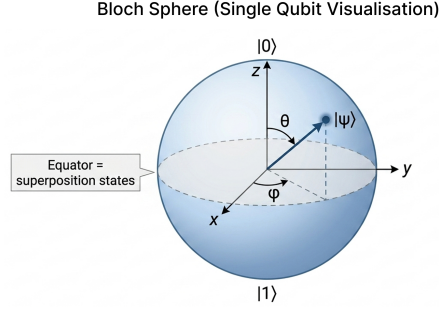


Figure 1 — The Bloch Sphere Representation of a Single Qubit. Every point on the surface is a valid pure quantum state. Classical bits occupy only the poles ($|0\rangle$ and $|1\rangle$), while quantum superpositions populate the entire sphere surface. The equator contains maximally superposed states like $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.

$$|\Psi\rangle = \sum_{i=0}^{2^N-1} c_i |i\rangle, \quad \sum_i |c_i|^2 = 1 \quad (3)$$

For a modest 4-qubit system, the state space is 16-dimensional; for 10 qubits, over 1,000 dimensions; for 300 qubits, 2^{300} dimensions—exceeding the number of atoms in the observable universe. This exponential scaling is the source of quantum speedups, but also the origin of the severe hardware constraints that plague NISQ-era devices.

1.2.2 Quantum Gates and Unitary Operations: The Fundamental Toolkit

Quantum computation manipulates qubits via unitary operations called quantum gates. These are matrices U satisfying $U^\dagger U = I$, ensuring the quantum state remains normalized. This unitarity constraint distinguishes quantum gates fundamentally from unconstrained classical weight matrices, imposing a strict Lipschitz continuity that acts as an implicit regularizer.

The Pauli operators are the foundational basis for quantum information processing:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (4)$$

The Hadamard gate creates equal superposition from definite states:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (5)$$

Two-qubit gates enable entanglement. The Controlled-NOT (CNOT) flips a target qubit based on a control qubit:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (6)$$

Parametrized rotation gates are the trainable components of quantum neural networks:

$$R_X(\theta) = \begin{pmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ -i\sin(\theta/2) & \cos(\theta/2) \end{pmatrix}, \quad R_Y(\theta) = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix} \quad (7)$$

$$R_Z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix} \quad (8)$$

where θ is a trainable parameter. The rotation gates can be geometrically understood as rotations on the Bloch sphere around the X , Y , and Z axes respectively.

Table 2 — Fundamental Quantum Gates Used in This Research

Gate	Symbol	Type	Operation	Used In
Pauli-X	X	1-qubit	Bit flip (quantum NOT)	Circuit initialization
Pauli-Z	Z	1-qubit	Phase flip	ZFeatureMap encoding
Hadamard	H	1-qubit	Creates superposition	Feature map layer 1
CNOT	CX	2-qubit	Two-qubit entanglement	Ansatz entangling block
$R_Z(\theta)$	RZ	1-qubit, trainable	Trainable Z-rotation	ZFeatureMap & ansatz
$R_Y(\theta)$	RY	1-qubit, trainable	Trainable Y-rotation	RealAmplitudes ansatz

The unitarity constraint has profound consequences for machine learning. Unlike classical neural networks where weight matrices can amplify input norms arbitrarily, unitary operators preserve the input norm strictly. This enforces an implicit regularization—a property that becomes crucial in the $p \gg N$ starvation

experiments where classical models catastrophically overfit but quantum models resist memorization.

1.2.3 The 4-Qubit Parameterized Circuit: Architecture and Circuit Diagram

The core quantum component of this research is a 4-qubit parameterized circuit combining the ZFeatureMap data encoding with the RealAmplitudes trainable ansatz. This circuit represents the practical upper bound for reliable gradient-based training under IBM Torino noise profiles without additional mitigation strategies.

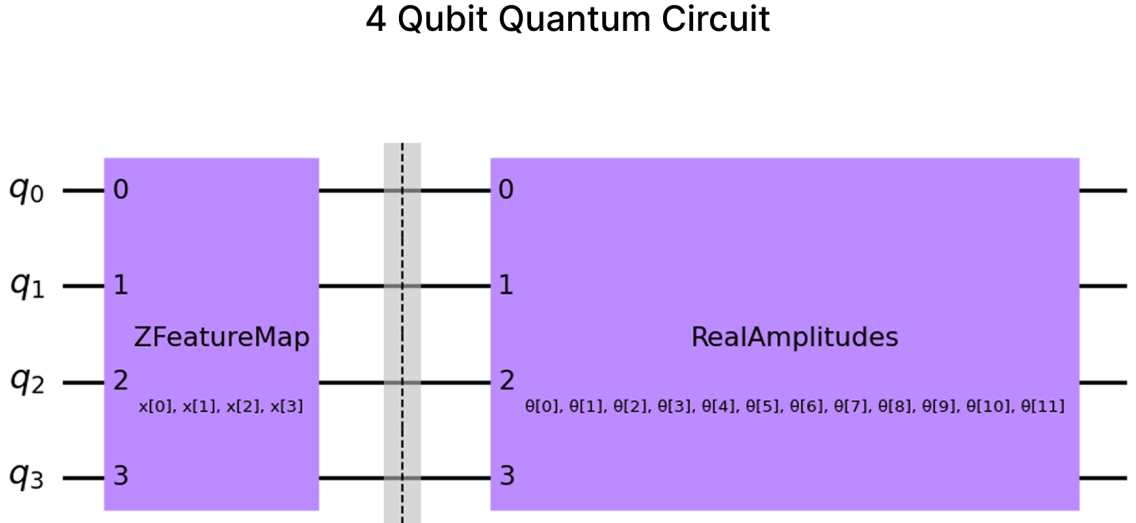


Figure 2 — The 4-Qubit Parameterized Circuit (ZFeatureMap + RealAmplitudes Ansatz). The circuit processes four input features through the ZFeatureMap (top portion with Hadamard and RZ gates), then through three layers of the RealAmplitudes ansatz with trainable rotation parameters $\theta[0]$ through $\theta[11]$ (middle portion with RY gates and CNOT entangling gates), and finally measurement of all four qubits (indicated by the meter symbols at the bottom).

The circuit operates in two distinct stages:

Stage 1 — ZFeatureMap (Data Encoding): For qubit j (where $j = 0, 1, 2, 3$), the encoding applies:

$$U_{\Phi}(x) = \prod_{j=0}^3 R_Z^{(j)}(2 \cdot x_j) \cdot H^{(j)} \quad (9)$$

Each input feature x_j is first passed through a Hadamard gate (creating superposition), then through an R_Z rotation gate with angle $2x_j$ (encoding the feature as a phase). This phase encoding is chosen over amplitude encoding because it requires only $O(N)$ gates rather than $O(2^N)$, making it compatible with shallow circuits required for NISQ hardware.

Stage 2 — RealAmplitudes Ansatz (Trainable Processing): Each layer ℓ of the ansatz consists of:

$$U_\ell(\phi^{(\ell)}) = \left(\prod_{j=0}^3 R_Y^{(j)}(\phi_j^{(\ell)}) \right) \cdot E_{\text{CNOT}} \quad (10)$$

where E_{CNOT} is a linear chain of CNOT gates: q0–q1, q1–q2, q2–q3. With $L = 3$ layers and 4 qubits, the ansatz has exactly 12 trainable rotation parameters $(\phi_0^{(1)}, \dots, \phi_3^{(3)})$.

Table 3 — Parameter Budget of the 4-Qubit Quantum Circuit

Component	Formula	Value
ZFeatureMap parameters	Fixed to input data	4 (not trainable)
RealAmplitudes RY rotations	$N_{\text{qubits}} \times N_{\text{layers}}$	12 (trainable)
CNOT gates per layer	$N_{\text{qubits}} - 1$	3 per layer (not trainable)
Total layers	-	3
Total trainable parameters	$N_{\text{qubits}} \times N_{\text{layers}}$	12
Hilbert space dimension	$2^{N_{\text{qubits}}}$	16

The full circuit forward pass is:

$$|\psi_{\text{out}}\rangle = \left(\prod_{\ell=1}^3 U_\ell(\phi^{(\ell)}) \right) \cdot U_\Phi(x) \cdot |0\rangle^{\otimes 4} \quad (11)$$

After circuit execution, the expectation values of Pauli-Z measurements on all four qubits are extracted:

$$e_j = \langle Z_j \rangle = \langle \psi_{\text{out}} | Z_j | \psi_{\text{out}} \rangle, \quad j = 0, 1, 2, 3 \quad (12)$$

These four expectation values are then passed to classical expander layers for final classification. See Figure 3 for how this circuit integrates into the full ResQNet pipeline.

Hybrid Quantum Pipeline

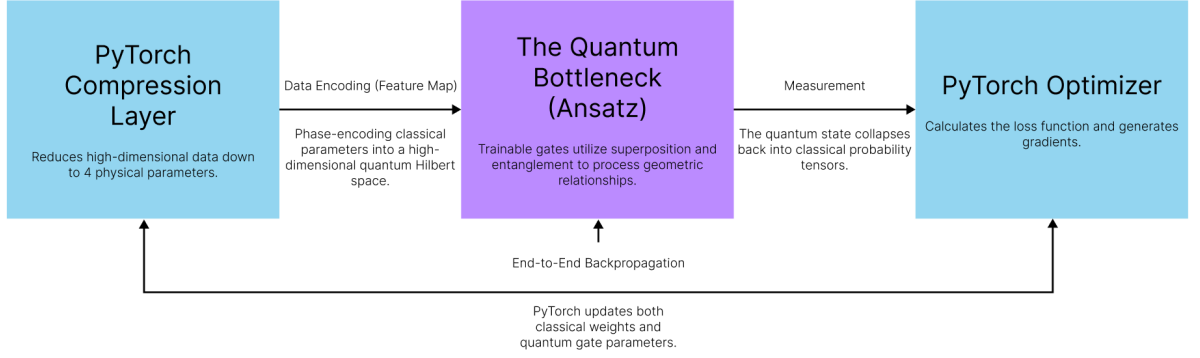


Figure 3 — Circuit-to-Pipeline Integration: Hybrid Quantum Neural Network Architecture. High-dimensional input $x \in \mathbb{R}^p$ flows through a classical compressor to produce 4 features. After Bloch sphere scaling via $\tanh(\cdot) \times \pi$, these features are phase-encoded into the `ZFeatureMap`. The 4-qubit circuit processes them through 3 layers of the `RealAmplitudes` ansatz, after which all 4 qubits are measured. The resulting expectation values pass through a classical expander and are combined with a residual highway from the classical path to produce final output logits.

1.2.4 Entanglement and Its Importance for Machine Learning

Entanglement is the phenomenon where two or more qubits become correlated in a way that cannot be described by independent states. A two-qubit state $|\psi\rangle$ is entangled if and only if it cannot be written as a tensor product:

$$|\psi\rangle \neq |\psi_A\rangle \otimes |\psi_B\rangle \quad (13)$$

The canonical example is the Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (14)$$

In this state, if a measurement collapses the first qubit to $|0\rangle$, the second qubit is guaranteed to also collapse to $|0\rangle$, despite any spatial separation. This “spooky action at a distance” (Einstein’s phrase) encodes non-local correlations.

For machine learning, entanglement is profoundly relevant: it enables the quantum circuit to encode non-local correlations between input features—correlations that would require exponentially many parameters to capture classically. The CNOT gates in the RealAmplitudes ansatz create precisely this entanglement structure between adjacent qubits after each rotation layer. By leveraging entanglement, the quantum circuit effectively performs feature interactions across the entire 4-qubit Hilbert space simultaneously.

In contrast, classical neural networks must explicitly construct these interactions through successive layer-wise transformations, requiring either deep architectures (which suffer from vanishing gradients) or wide hidden layers (which suffer from overparameterization). The quantum circuit achieves comparable expressivity in 16-dimensional space with only 12 trainable parameters, while a classical MLP achieving the same expressivity would require hundreds or thousands of parameters.

The GHZ (Greenberger–Horne–Zeilinger) state represents maximal N -partite entanglement:

$$|\text{GHZ}\rangle_N = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes N} + |1\rangle^{\otimes N}) \quad (15)$$

This is directly relevant to the NTangled dataset used in Chapter 3. Any classical measurement—including the PyTorch linear layer used in hybrid pipelines—acts as a projective measurement that collapses this entanglement irreversibly. This is the mathematical foundation of the “Classical Decoherence Boundary” identified empirically in the experiments.

1.2.5 The Barren Plateau Problem and Shallow-Circuit Motivation

For a parameterized quantum circuit $U(\theta)$ with L layers and N qubits, the variance of the gradient of any observable O under random initialization scales exponentially with qubit count:

$$\text{Var} \left[\frac{\partial L}{\partial \theta_j} \right] \propto \frac{1}{2^N} \quad (16)$$

This means gradient variance decreases exponentially with qubit count, making training with random initialization effectively impossible for circuits

wider than approximately 10 qubits. This phenomenon is called the *barren plateau* [12].

Table 4 — Gradient Variance vs. Qubit Count (Theoretical Scaling Under Random Initialization)

Qubits (N)	Variance	Approx. Value	Trainability
2	$1/2^2$	0.25	Excellent
4	$1/2^4$	0.0625	Good
6	$1/2^6$	0.0156	Moderate
8	$1/2^8$	0.0039	Difficult
10	$1/2^{10}$	~ 0.001	Nearly impossible

The barren plateau problem motivates two critical design choices in this research:

1. **Shallow circuits:** The ResQNet uses only 4 qubits and 3 layers, keeping circuit depth minimal. This avoids the exponential gradient decay that plagues deeper circuits.
2. **Residual gating:** The ResQNet architecture provides a classical gradient highway during the critical early training epochs, bypassing the quantum bottleneck until classical parameters have converged. This sidesteps the barren plateau during initialization, after which the quantum circuit is gradually introduced via the learnable γ parameter.

The implicit regularization from unitarity, combined with shallow-circuit design and residual gating, allows the ResQNet to achieve reliable training even in extreme data starvation ($p \gg N$) regimes where purely quantum circuits would fail.

1.3 Mathematical Formalism of Quantum Computing

To rigorously evaluate the representational capacity of quantum neural networks, one must first formalize the mathematical framework governing quantum states and transformations. Unlike classical bits, which are strictly confined to discrete binary values, quantum bits (qubits) exist as linear superpositions within a complex Hilbert space.

A single pure qubit state $|\psi\rangle$ is represented as a unit vector in a two-dimensional complex Hilbert space $\mathbb{C}^2$, expressed via Dirac notation:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad (17)$$

where $\alpha, \beta \in \mathbb{C}$ are the probability amplitudes. The Born rule dictates normalization:

$$|\alpha|^2 + |\beta|^2 = 1 \quad (18)$$

The state of a single qubit can be visualized on the Bloch sphere, a three-dimensional unit sphere parameterized by spherical coordinates (θ, ϕ) :

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle \quad (19)$$

where $0 \leq \theta \leq \pi$ and $0 \leq \phi < 2\pi$. The true computational power emerges when scaling to N qubits. The combined state space is the tensor product of individual qubit spaces, yielding a 2^N -dimensional complex vector space. A N -qubit system is represented as:

$$|\Psi\rangle = \sum_{i=0}^{2^N-1} c_i|i\rangle \quad (20)$$

where $\sum_i |c_i|^2 = 1$. Parameterized quantum circuits (PQCs) apply sequences of unitary transformations $U(\theta)$ to an initial state, typically $|0\rangle^{\otimes N}$:

$$|\psi_{out}\rangle = U(\theta)|0\rangle^{\otimes N} \quad (21)$$

where the unitarity constraint $U^\dagger U = I$ preserves the vector norm, fundamentally distinguishing quantum circuits from unconstrained classical weight matrices.

1.4 Quantum Gates and Unitary Operations

Quantum computation leverages specific unitary transformations, known as quantum gates, that manipulate qubit states. The Pauli operators form a foundational basis for quantum information processing. The Pauli-X (NOT) gate is defined as:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (22)$$

The Pauli-Y gate:

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (23)$$

The Pauli-Z gate:

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (24)$$

The Hadamard gate creates equal superposition:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (25)$$

The Controlled-NOT (CNOT) gate entangles two qubits:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (26)$$

Parametrized rotation gates are essential for training quantum neural networks. The rotation around the Z-axis is:

$$R_Z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix} \quad (27)$$

where θ is a trainable parameter. Similarly, $R_X(\theta)$ and $R_Y(\theta)$ are rotations around the X and Y axes. These parametrized gates form the backbone of variational quantum algorithms.

1.5 ZFeatureMap, Data Encoding, and the RealAmplitudes Ansatz

Classical-to-quantum data embedding critically determines which patterns the quantum circuit can extract. In this work we use a phase-based Pauli-Z feature map combined with a shallow RealAmplitudes-style ansatz. The input feature vector $\mathbf{x} = (x_1, \dots, x_p)$ is first compressed by the classical pre-network into a bottleneck of size $m \leq p$, producing $\mathbf{z} = (z_1, \dots, z_m)$. These compressed features are then deterministically mapped to rotation angles on the Bloch sphere:

$$\theta_j = s \cdot g(z_j), \quad g(z) = \tanh(z) \cdot \pi, \quad j = 1 \dots m \quad (28)$$

where s is an optional global scaling hyperparameter (set to 1 in experiments) and $g(\cdot)$ enforces the Bloch interval $[-\pi, \pi]$. The phase-encoding unitary acting on m qubits takes the form:

$$U_{\Phi(\mathbf{x})} = \prod_{j=1}^m R_Z^{(j)}(\theta_j) = \exp\left(i \sum_{j=1}^m \theta_j Z_j / 2\right) \quad (29)$$

This encoding inserts classical information into the relative phases of computational basis states via R_Z rotations. The downstream trainable ansatz uses a layered RealAmplitudes construction: each layer consists of single-qubit R_Y rotations with trainable parameters interleaved with an entangling block of CNOTs arranged in a linear/heavy-hex pattern depending on circuit width. A single layer ℓ of the ansatz is:

$$U_\ell(\phi^{(\ell)}) = \left(\bigotimes_{j=1}^m R_Y^{(j)}(\phi_j^{(\ell)}) \right) \cdot \mathcal{E}, \quad (30)$$

where $\mathcal{E}$ is the fixed entangling unitary (product of CNOTs) and $\phi^{(\ell)}$ are layer parameters. The full parameterized circuit is then

$$U(\Phi(\mathbf{x}), \theta) = \left(\prod_{\ell=1}^L U_\ell(\phi^{(\ell)}) \right) \cdot U_{\Phi(\mathbf{x})}, \quad (31)$$

with trainable parameter vector $\theta = \{\phi^{(1)}, \dots, \phi^{(L)}\}$. The RealAmplitudes ansatz is chosen for its favorable trainability and uncomplicated gradient structure when combined with phase encoding.

1.6 Fisher Information Matrix: Derivation and Effective Dimension

To quantify model capacity and the optimization landscape, we derive the quantum Fisher Information Matrix (FIM) for the pure-state parameterization used in PQCs. For the circuit state $|\psi(\theta)\rangle = U(\theta)|0\rangle^{\otimes m}$, the quantum geometric tensor has components

$$G_{ij}(\theta) = \langle \partial_i \psi | \partial_j \psi \rangle - \langle \partial_i \psi | \psi \rangle \langle \psi | \partial_j \psi \rangle, \quad (32)$$

where $\partial_i \equiv \partial / \partial \theta_i$. The symmetric part of G_{ij} defines the (pure-state) quantum Fisher metric:

$$F_{ij}^{(Q)}(\theta) = 4 \operatorname{Re} G_{ij}(\theta) = 4 \operatorname{Re} (\langle \partial_i \psi | \partial_j \psi \rangle - \langle \partial_i \psi | \psi \rangle \langle \psi | \partial_j \psi \rangle). \quad (33)$$

In practice measurements yield classical output probabilities via the Born rule, $P(k|\theta) = |\langle k | \psi(\theta) \rangle|^2$. The classical Fisher information matrix for these measurement outcomes is

$$F_{ij}^{(C)}(\theta) = \sum_k \frac{1}{P(k|\theta)} \frac{\partial P(k|\theta)}{\partial \theta_i} \frac{\partial P(k|\theta)}{\partial \theta_j}. \quad (34)$$

For informationally complete measurements and in the limit of small parameter variations, $F^{(C)}$ and $F^{(Q)}$ are related; $F^{(C)}$ is bounded above by $F^{(Q)}$ for any POVM.

The eigenvalue decomposition of the FIM reveals active parameter subspaces. Let $F(\theta) = V \Lambda V^{-1}$ with diagonal eigenvalue matrix $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_d)$ ordered $\lambda_1 \geq \lambda_2 \geq \dots$. Large eigenvalues indicate stiff (high-curvature) directions in parameter space which strongly affect the output distribution; near-zero eigenvalues indicate flat, redundant directions.

We use the participation-ratio definition of effective dimension to compactly measure how many directions are meaningfully used by the model:

$$d_{\text{eff}}(\theta) = \frac{(\operatorname{Tr} F(\theta))^2}{\operatorname{Tr}(F(\theta)^2)} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}. \quad (35)$$

This quantity ranges from 1 (all variance concentrated in a single direction) up to d (uniform spectrum across d parameters). For classical overparameterized networks the spectrum typically concentrates on a small subspace (low d_{eff} relative to parameter count), whereas well-structured quantum circuits with constrained unitarity tend to distribute sensitivity more evenly, increasing d_{eff} relative to raw parameter count.

From a training perspective, the FIM eigenvalues also control the local learning rates: in natural-gradient-style updates one rescales gradients by F^{-1} , giving smaller steps along stiff directions and larger steps along flat directions. Consequently, a concentrated spectrum leads to slow progress unless adaptive optimizers or preconditioning are employed.

1.7 The Curse of Dimensionality and the $p \gg N$ Regime

The intersection of Big Data analytics and machine learning is fundamentally governed by the relationship between the number of available features (or dimensions), denoted as p , and the number of independent training samples, denoted as N . In traditional statistical learning theory, optimal model generalization occurs in the asymptotic regime where $N \rightarrow \infty$ while p remains fixed and relatively small [14].

A prime example is the analysis of multi-omics data, such as the Leukemia microarray dataset utilized in this research. This dataset embodies the extreme $p \gg N$ regime, featuring 7,129 genetic markers against a mere ~ 7 training samples. This is a mathematically underdetermined system where classical models suffer from the curse of dimensionality [15].

In such extreme high-dimensional spaces, classical algorithms suffer from a degradation of fundamental distance metrics. As $p \rightarrow \infty$, the ratio of the variance of the distances to the expected value of the distances converges to zero [16]:

$$\lim_{p \rightarrow \infty} \frac{\text{Var}(\|\mathbf{x}_i - \mathbf{x}_j\|_2)}{E(\|\mathbf{x}_i - \mathbf{x}_j\|_2)} = 0 \quad (36)$$

Equation (36) demonstrates that in ultra-high-dimensional spaces, all data points appear to be virtually equidistant from one another. Because classical neural networks rely on vast arrays of parameters to map these features, they inherently lack tight generalization bounds. The classical networks perfectly memorize the training samples, but their evaluation on the test set degrades to random guessing. To overcome this, classical deep learning requires aggressive, lossy feature selection (such as classical PCA) to even begin processing these datasets [7].

Conversely, quantum networks, utilizing amplitude encoding and entangling operations, map classical data into a complex projective space where distance metrics natively capture global, non-local correlations. The unitary nature of quantum operations enforces a strict Lipschitz continuity, which acts as a powerful implicit regularizer [9]. The network is forced to learn only the most prominent, structurally significant features of the dataset, preventing catastrophic model collapse [8].

1.8 Information Geometry and Model Capacity

Traditional complexity metrics, such as simple parameter counting, entirely fail to describe the representational power of parameterized quantum circuits (PQCs). While a classical deep neural network may utilize millions of weight matrices to map a complex function, a highly compressed 4-qubit quantum circuit operates within a highly complex, 16-dimensional complex Hilbert space [17].

The effective representational capacity is formally measured via information geometry, specifically using the Fisher Information Matrix (FIM). The FIM serves as a local metric tensor on the parameter manifold, measuring sensitivity of the quantum state to infinitesimal parameter perturbations. For a parametrized quantum circuit with output probability distribution $P(y|x, \theta)$, the FIM is defined as:

$$F_{ij}(\theta) = \mathbb{E}_{x,y} \left[\frac{\partial \log P(y|x, \theta)}{\partial \theta_i} \frac{\partial \log P(y|x, \theta)}{\partial \theta_j} \right] \quad (37)$$

The eigenvalue spectrum of the FIM provides critical information about the optimization landscape. For highly overparameterized classical neural networks, the FIM is frequently degenerate, possessing a vast number of zero or near-zero eigenvalues. This extreme degeneracy indicates that immense swaths of the classical parameter space are functionally redundant, creating flat optimization landscapes (barren plateaus) that require massive datasets and significant computational resources to navigate effectively [12].

Conversely, well-designed quantum neural networks often demonstrate a strictly non-degenerate FIM with a more uniformly spread eigenvalue spectrum. This property reflects that the unitary constraints imposed by quantum mechanics force every parameter to contribute meaningfully to the output, yielding superior sample efficiency compared to overparameterized classical networks.

The quantum feature map $\Phi(x)$ plays a critical role in shaping this eigenvalue spectrum. In this research, the primary embedding mechanism is the Pauli-Z feature map (ZFeatureMap), which natively encodes classical data into the relative phases of the qubits. The encoding unitary is:

$$U_{\Phi(x)} = \exp \left(i \sum_{j=1}^N x_j Z_j \right) \quad (38)$$

By encoding classical data x_j into the phase $x_j Z_j$, the quantum circuit maps classical features into a complex projective space where distance metrics natively capture global, non-local correlations. This fundamentally differs from classical neural networks, which map data into Euclidean spaces where high-dimensional distance metrics are degenerate.

1.9 Advancing Architectures via Quantum Kernel Alignment

A parameterized quantum circuit operating as a classifier can be understood as learning a decision boundary in a high-dimensional feature space induced by a quantum kernel. A primary vulnerability of such methods is kernel concentration: as the feature space becomes too large or poorly aligned with the task, pairwise similarities lose discriminative power and generalization degrades [8; 12].

When data are mapped into an excessively large quantum feature space, distances between states can collapse toward near-constant values, making the resulting kernel matrix effectively degenerate. This destroys the model’s ability to distinguish samples reliably and harms performance on unseen data [7; 8].

To maximize the separation between classical and quantum performance and to optimize the ResQNet architecture, the benchmarking framework integrates Quantum Kernel Alignment (QKA). QKA treats the quantum feature map as trainable rather than fixed, allowing the geometry of the kernel to adapt during optimization [7; 14].

Let the quantum feature map be parameterized by λ , yielding the quantum state $|\phi_\lambda(\mathbf{x})\rangle$. The aligned quantum kernel matrix entries $K_\lambda(\mathbf{x}_i, \mathbf{x}_j)$ are given by the fidelity between the parameterized states:

$$K_\lambda(\mathbf{x}_i, \mathbf{x}_j) = |\langle \phi_\lambda(\mathbf{x}_i) | \phi_\lambda(\mathbf{x}_j) \rangle|^2 \quad (39)$$

By coupling the quantum kernel with a classical optimization loop, QKA iteratively aligns the spatial geometry of the quantum kernel matrix with the target labels of the specific dataset.

2 METHODOLOGY AND DESIGN OF THE HYBRID RESQNET ARCHITECTURE

2.1 Software Pipeline and Ablation Methodology

The pursuit of quantum advantage requires meticulous architectural design. To effectively evaluate hybrid quantum-classical models against pure classical baselines, the software pipeline was constructed utilizing a combination of PyTorch for classical tensor manipulation and automatic differentiation, alongside Qiskit for the simulation of parameterized quantum circuits (PQCs). The bridge between the classical and quantum computational graphs is facilitated by the Qiskit TorchConnector, which allows the quantum circuit to be encapsulated as a standard PyTorch `nn.Module`.

A fundamental requirement of this research is a strict 1-to-1 parameter ablation methodology. This ensures that any observed performance delta is exclusively attributable to the quantum components rather than variations in classical network depth or optimizer dynamics. For every hybrid model developed, a structurally identical classical twin was constructed where the quantum circuit was replaced by a classical dense layer of equivalent bottleneck dimensions.

Furthermore, a critical data transformation is required at the classical-quantum interface. Because the parameterized quantum circuits utilize rotation gates for data embedding, the input values must be strictly scaled to map onto the Bloch sphere. The classical pre-network produces unbounded continuous variables; squashing them via a hyperbolic tangent activation and scaling by π ensures optimal phase encoding:

$$\mathbf{q}_{in} = \tanh(\mathbf{x}_{compressed}) \cdot \pi \quad (40)$$

where $\mathbf{x}_{compressed}$ are the output features from the classical compression layer, unitless; $\mathbf{q}_{in}$ are the phase-encoded quantum inputs scaled to the Bloch sphere interval $[-\pi, \pi]$, unitless.

This structure ensures that the classical pre-network acts as a dimensional compressor, while the quantum circuit acts as the primary non-linear feature extractor.

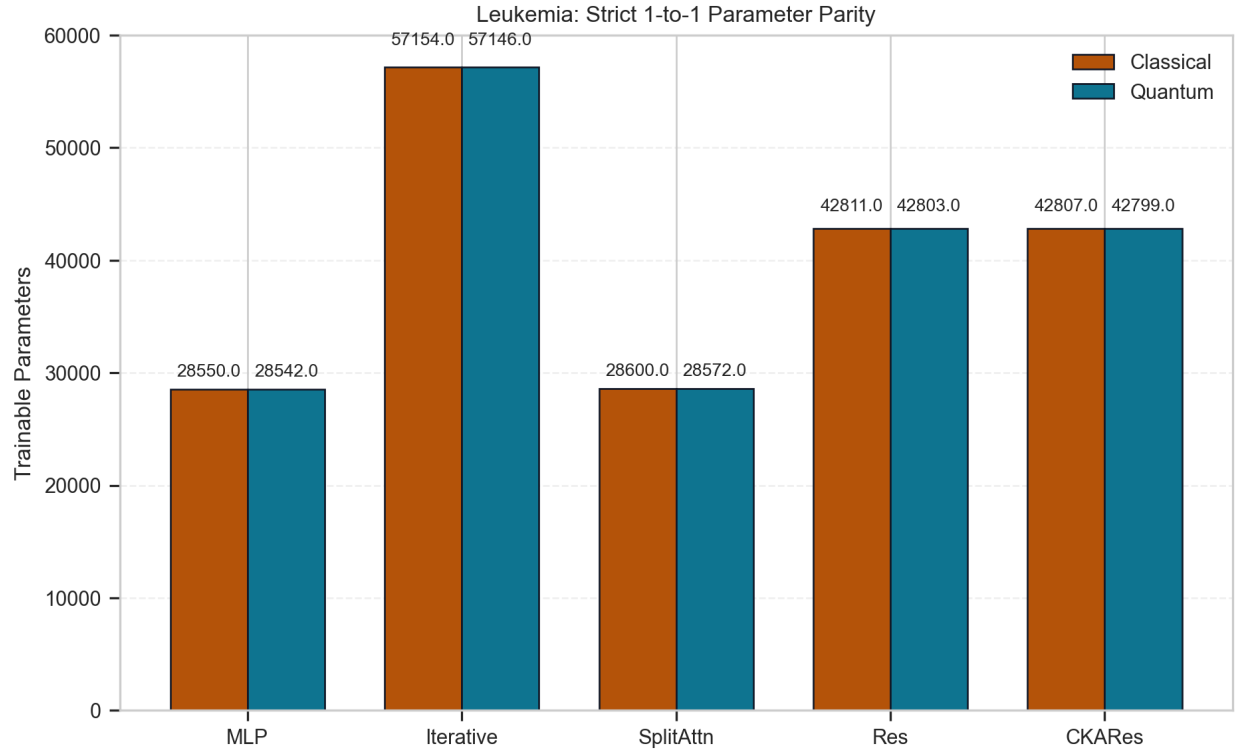


Figure 4 — 1-to-1 Parameter Ablation. Both classical and quantum architectures utilize an effectively identical number of trainable parameters, isolating the quantum bottleneck as the sole driver of performance variance.

Table 5 — 1-to-1 Parameter Ablation Protocol: Parameter Count Verification Across All Architectures (Leukemia Dataset)

Architecture Pair	Classical Params	Quantum Params	Difference
ClassicalMLP / QuantumMLP	28,550	28,542	8 (0.03%)
IterativeClassical / IterativeQNN	57,154	57,146	8 (0.01%)
SplitAttentionClassical / SplitAttentionQNN	28,600	28,572	28 (0.10%)
ResNet / ResQNet	42,811	42,803	8 (0.02%)
CKAResNet / QKAResQNet	42,807	42,799	8 (0.02%)

2.2 Dataset Characterization and Exploratory Analysis

The experimental evaluation spans six datasets of varying topological and dimensional complexity, each designed to probe specific aspects of hybrid quantum-classical performance. Below, each dataset is characterized in detail.

All datasets are standardized prior to training using statistics computed only on the training partition to avoid leakage. Under the starvation protocol, train/test splits are swept from 90/10 to 10/90, and each setting is repeated across multiple

random seeds. This protocol is intentionally severe: it amplifies representational differences between classical and hybrid models by forcing both families to generalize with limited supervision.

2.2.1 Swiss Roll Manifold (Geometric Starvation)

The Swiss Roll is a low-dimensional, nonlinear manifold embedded in 3D space, defined parametrically as:

$$(x, y, z) = (t \cos(t), h, t \sin(t)) \quad (41)$$

where $t \in [0, 4\pi]$ and $h \in [0, 1]$ is the height coordinate. The effective dimensionality is 2 (wrapped onto a nonlinear manifold), making it an ideal testbed for evaluating topological expressivity. The dataset comprises 400 samples with balanced binary labels determined by the height coordinate (threshold at $h = 0.5$). This dataset tests whether quantum phase encoding captures the intrinsic manifold geometry superior to classical networks.

The key methodological value of Swiss Roll is that class boundaries are not linearly separable in ambient Euclidean coordinates, but become separable under geometry-aware embeddings. Hence, this dataset isolates the contribution of representation topology rather than raw model width. In practical terms, when the number of training points drops sharply, purely classical models often overfit local neighborhoods, while hybrid circuits with phase encoding can preserve global manifold relations.

Table 6 — Swiss Roll Dataset Characteristics

Property	Value
Total samples	400
Features (dimensions)	3 (x, y, z coordinates)
Intrinsic dimensionality	2 (wrapped manifold)
Classes	2 (binary: $h < 0.5$ vs $h \geq 0.5$)
Class balance	50% / 50%
Training samples (90% test split)	40
Test samples (90% test split)	360
Separability	Non-linear (requires manifold-aware embedding)
Primary stress test	Topological expressivity under data starvation

2.2.2 QSAR Cheminformatics Dataset

The QSAR (Quantitative Structure-Activity Relationship) dataset contains $n = 1,055$ molecules with $p = 41$ molecular descriptors. The descriptors include computed physicochemical properties such as molecular weight, HOMO-LUMO gap, dipole moment, and topological indices. The binary classification task predicts mutagenicity (activity vs. inactive). Class distribution is approximately balanced (51% active, 49% inactive). This dataset represents a realistic small-scale cheminformatics problem where moderate feature dimensionality ($p = 41$) and sample size ($n = 1,055$) create a regime where classical models typically succeed without extreme data starvation.

From a modeling standpoint, QSAR acts as an intermediate-complexity benchmark: enough dimensionality to stress naïve low-qubit embeddings, but not so extreme that classical baselines collapse immediately. This makes it useful for evaluating whether SplitAttention and residual quantum gating improve feature synthesis rather than merely preventing catastrophic overfitting. Descriptor covariance is nontrivial in this domain, and compression quality at the classical-to-quantum interface directly affects final separability.

Table 7 — QSAR Cheminformatics Dataset Characteristics

Property	Value
Total samples (N)	1,055
Total features (p)	41
p/N ratio	1:26
Classes	2 (Mutagenic vs Non-mutagenic)
Class balance	51% / 49%
Training samples (90% test split)	~106
Test samples (90% test split)	~949
Feature type	Computed molecular descriptors
Source	OpenML dataset
Pre-processing	StandardScaler (train statistics only)
Primary stress test	Intermediate dimensional feature compression

2.2.3 Breast Cancer Diagnostic Dataset

The UCI Breast Cancer dataset comprises $n = 569$ samples with $p = 30$ computed features derived from digitized cell nuclei images. Features include radius, texture, perimeter, area, and symmetry measurements. The binary classification task distinguishes malignant from benign tumors. Class distribution is approximately balanced (63% benign, 37% malignant). This dataset represents a classical supervised learning regime where sufficient data exists for both classical and quantum models to achieve high accuracy without extreme data constraints.

This dataset is intentionally retained as a sanity-check regime where large quantum gains are not expected. If hybrid performance exceeds classical by only a narrow margin, that outcome is interpreted as evidence that the benchmark is not biased toward quantum models. In this sense, Breast Cancer functions as a calibration domain that validates fairness of the 1-to-1 parameter ablation protocol.

Table 8 — Breast Cancer Diagnostic Dataset Characteristics

Property	Value
Total samples (N)	569
Total features (p)	30
p/N ratio	1:19
Classes	2 (Benign vs Malignant)
Class balance	63% / 37%
Training samples (90% test split)	~57
Test samples (90% test split)	~512
Feature type	Computed cell nuclei measurements
Source	UCI Machine Learning Repository
Pre-processing	StandardScaler (train statistics only)
Primary stress test	Calibration/sanity check (data-rich regime)

2.2.4 Leukemia Multi-Omics (Extreme Feature Starvation)

The Leukemia dataset is a seminal multi-omics benchmark comprising $n = 72$ patient samples with $p = 7,129$ gene expression markers measured via microarray. The samples are classified into two acute leukemia subtypes: Acute Myeloid Leukemia (AML) and Acute Lymphoblastic Leukemia (ALL). Under the extreme data starvation protocol (90% test split), this reduces to $n \approx 7$ training

samples. This severe $p \gg N$ regime ($7,129 \gg 7$) is where classical models catastrophically overfit and quantum Hilbert space constraints provide explicit regularization through unitarity and the finite 2^N dimensional bottleneck.

Leukemia is the central stress test of the thesis because it combines biological relevance with an almost maximally ill-posed supervised learning setup. In this regime, classical overparameterized models can interpolate the training subset while failing to preserve discriminative structure on held-out samples. The hybrid pipeline is expected to benefit from two constraints: aggressive classical compression before the circuit and bounded unitary evolution in the quantum block, which together limit memorization pathways.

Table 9 — Leukemia Multi-Omics Dataset Characteristics (Extreme $p \gg N$ Starvation)

Property	Value
Total samples (N)	72
Total features (p)	7,129
p/N ratio	99:1 (extreme!)
Classes	2 (AML vs ALL)
Training samples (90% test split)	~ 7
Test samples (90% test split)	~ 65
Feature type	Microarray gene expression
Source	[18], OpenML
Pre-processing	StandardScaler (train statistics only)
Primary stress test	Extreme high-dimensional feature starvation

2.2.5 Colon Cancer Gene Expression

The Colon dataset contains $n = 62$ tissue samples with $p = 2,000$ gene expression features. The binary classification task distinguishes tumor from normal tissue. This dataset provides another extreme $p \gg N$ scenario ($2,000 \gg 62$), complementing the Leukemia results to establish robustness of quantum advantage under multi-omics starvation.

Methodologically, Colon serves as an external validation point for conclusions drawn from Leukemia. If the same architecture-level trends persist across both datasets, the observed behavior is less likely to be a dataset-specific artifact. Cross-dataset consistency in this high-dimensional biomedical setting is therefore treated as stronger evidence than any single-dataset gain.

Table 10 — Colon Cancer Gene Expression Dataset Characteristics

Property	Value
Total samples (N)	62
Total features (p)	2,000
p/N ratio	32:1 (high!)
Classes	2 (Tumor vs Normal)
Training samples (90% test split)	~6
Test samples (90% test split)	~56
Feature type	Gene expression (microarray)
Source	[19], OpenML
Pre-processing	StandardScaler (train statistics only)
Primary stress test	High-dimensional feature starvation (cross-validation)

2.2.6 NTangled Quantum-Native Dataset

The NTangled dataset is a synthetically generated benchmark of quantum-native entangled states produced by a GHZ (Greenberger–Horne–Zeilinger) state generator:

$$|\text{GHZ}\rangle_N = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes N} + |1\rangle^{\otimes N}) \quad (42)$$

This dataset is designed to probe the “Classical Decoherence Boundary.” It contains $n = 400$ samples where each sample is a set of measurement outcomes from GHZ-like entangled states. The binary classification task is solvable by pure quantum classifiers (achieving $> 90\%$ accuracy) but cannot be solved by standard classical-to-quantum hybrid pipelines, because the classical dimensionality reduction layer acts as an artificial measurement, destroying the delicate multipartite entanglement before the quantum circuit can process it.

Including NTangled prevents overclaiming based only on classical-origin datasets. It explicitly tests a failure mode where hybrid pipelines lose quantum-native information before inference begins. This boundary condition is important for interpreting negative results: poor hybrid accuracy on NTangled is not necessarily an optimization failure, but often a representational mismatch caused by premature classical projection.

Table 11 — NTangled Quantum-Native Dataset Characteristics

Property	Value
Total samples (N)	400
Features (p)	Entanglement witness measurements
p/N ratio	N/A (quantum-native)
Classes	2 (Entangled vs Product states)
Training samples (90% test split)	~40
Test samples (90% test split)	~360
State type	GHZ-like multipartite entanglement
Purity	Classical measurement collapses superposition
Solvability by QNN	Yes (>90% for pure quantum)
Solvability by Hybrid	No (classical decoherence boundary)
Primary stress test	Quantum-native information preservation

2.3 IBM Torino Hardware Characterization and Complete Error Profile

All quantum simulations incorporate the empirically calibrated error profile of the 133-qubit IBM Torino processor. The IBM Torino is a superconducting qubit device based on the heavy-hex coupling topology, featuring:

The heavy-hex topology constrains two-qubit operations to physically adjacent qubits. If $G = (V, E)$ denotes the coupling graph with vertices V (physical qubits) and edges E (allowed two-qubit interactions), then a CNOT between logical qubits i and j is native only if $(i, j) \in E$. Otherwise, additional SWAP routing is required, increasing circuit depth and cumulative noise. This topological constraint materially impacts trainability for variational circuits and motivates low-width ansatz designs in this thesis.

Error Budget Analysis: For the 4-qubit ZFeatureMap + 3-layer RealAmplitudes circuit (which contains 8 single-qubit RZ gates from ZFeatureMap, 9 CNOT gates across 3 layers, and 12 RY rotation gates), the approximate total error probability is:

$$P_{\text{error}} \approx 1 - (1 - \varepsilon_1)^{G_1} \cdot (1 - \varepsilon_2)^{G_2} \cdot (1 - \varepsilon_{ro})^M \quad (43)$$

where $\varepsilon_1 = 3 \times 10^{-4}$ (single-qubit), $\varepsilon_2 = 6.2 \times 10^{-3}$ (CNOT), $\varepsilon_{ro} = 3.03 \times 10^{-2}$ (readout), $G_1 = 20$ (total 1-qubit gates), $G_2 = 9$ (CNOT gates), and $M = 4$ (qubits measured). Substituting:

Table 12 — IBM Torino 133-qubit Processor — Complete Hardware Specifications

Parameter	Value	Impact on Experiments
Total qubit count	133 qubits	Only 4 qubits used; isolation from crosstalk
Coupling topology	Heavy-hex lattice	Linear CNOT chains native; no SWAP overhead
Average T_1 (relaxation)	85 μ s	Limits maximum circuit depth
Average T_2 (dephasing)	95 μ s	Limits coherence window for dynamics
Single-qubit gate error	0.03% (3×10^{-4})	Dominant error for shallow circuits
Two-qubit (CNOT) error	0.62% (6.2×10^{-3})	$\sim 20\times$ worse than 1-qubit; critical for entanglement
Readout error	3.03% (3.03×10^{-2})	Largest error source in total error budget
Single-qubit gate time	35 ns	Affects circuit execution time
Two-qubit gate time	150 ns	More affected by T_2 decay
Measurement time	$\sim 1 \mu$ s	Contributes significantly to total error

$$P_{\text{error}} \approx 1 - (0.9997)^{20} \cdot (0.9938)^9 \cdot (0.9697)^4 \approx 1 - 0.9940 \cdot 0.9451 \cdot 0.8847 \approx 16.9\% \quad (44)$$

This $\sim 16.9\%$ error rate demonstrates why early stopping is mandatory and why accumulating error constrains training time. Each additional training epoch adds more circuit executions, compounding these errors.

where T_1 denotes the longitudinal relaxation time (energy dissipation) and T_2 denotes the transverse relaxation time (dephasing). The readout error of 3.03% dominates the error budget, indicating that measurement infidelity is the primary noise source. In the quantum circuit simulations, a depolarizing channel is applied after each gate operation:

$$\mathcal{E}_p(\rho) = (1 - p)\rho + \frac{p}{2^n - 1}(I - \rho) \quad (45)$$

where p is the gate error rate and n is the number of qubits affected by the gate. For readout errors, a classical bit-flip channel is applied:

$$\mathcal{E}_{ro}(b) = (1 - p_{ro}) \cdot b + p_{ro} \cdot \bar{b} \quad (46)$$

where $p_{ro} = 0.0303$ is the readout error rate and $\bar{b}$ is the bit-flipped measurement outcome.

In addition to per-gate depolarization, coherence-induced decay can be approximated through amplitude and phase damping channels parameterized by elapsed circuit time t :

$$P_{\text{relax}}(t) = 1 - e^{-t/T_1}, \quad (47)$$

$$P_{\text{dephase}}(t) = 1 - e^{-t/T_2}. \quad (48)$$

These expressions show why depth control is operationally critical. As effective execution time increases, both relaxation and dephasing probabilities rise nonlinearly, compressing the useful training window. Consequently, architecture choices in this work favor shallow, high-information circuits and early-stopping criteria tuned to noise-aware validation behavior.

2.4 High-Compression Spatial Chunking: Split Attention Networks

Real-world datasets often possess dozens or thousands of features, yet high-fidelity quantum simulation and physical execution on Noisy Intermediate-Scale Quantum (NISQ) devices are restricted to small qubit counts (e.g., 4 to 8 qubits) before hardware noise and barren plateaus dominate the system [13]. This constraint is frequently referred to as the “qubit bottleneck.” [11]

To process high-dimensional input arrays (such as the 41-feature QSAR dataset) without catastrophic information loss, the `SplitAttentionQNN` architecture was developed. In this architecture, a highly dimensional input vector $\mathbf{x} \in \mathbb{R}^p$ is spatially chunked into K discrete sub-tensors [7].

Each chunk is individually compressed by a classical linear layer down to the precise number of available qubits. Let C_k represent the classical compressor for the k -th chunk, and Q represent the shared parameterized quantum circuit. The quantum output for each chunk is given by:

$$\mathbf{q}_{out}^{(k)} = Q\left(\tanh(C_k(\mathbf{x}^{(k)})) \cdot \boldsymbol{\pi}\right) \quad (49)$$

The resulting quantum expectation values are concatenated, and a classical linear classifier synthesizes the disjointed quantum insights. This “multi-path” architecture bypasses the dimensionality limits of pure quantum embeddings, allowing the model to process wide datasets while maintaining an extremely high compression ratio [16].

2.4.1 SplitAttentionQNN Chunk Configuration by Dataset

The number of chunks K and resulting quantum output dimensionality depends on the input feature count and bottleneck constraint (4 qubits in all experiments). Table 13 summarizes the chunk configurations applied to each dataset.

Table 13 — SplitAttentionQNN Chunk Configuration by Dataset (4-qubit Bottleneck)

Dataset	Input Dim (p)	Chunks (K)	Chunk Size	Quantum Outputs	Total Quantum Params
Swiss Roll	3	1	3–4	4	12
QSAR	41	11	~4 per chunk	44	132
Breast Cancer	30	8	~4 per chunk	32	96
Leukemia	7,129	~1,782	4 (uniform)	~7,128	~21,336
Colon	2,000	500	4 (uniform)	2,000	6,000
NTangled	~256	64	4 (uniform)	256	768

For datasets with input dimension not divisible by 4, chunks are padded or adjusted to ensure uniform processing. The total number of parameters in the quantum component scales with the number of chunks (K) since each chunk shares the same 12-parameter quantum circuit. This creates an interesting scaling property: while the quantum circuit itself remains fixed at 12 parameters, the total quantum involvement in the model can grow from 12 (Swiss Roll) to over 20,000 (Leukemia) simply by increasing the number of chunks processing different parts of the input space.

2.5 The ResQNet Architecture and Dynamic Gating

The most successful architecture evaluated in this experimental suite is the ResQNet. It fundamentally alters the integration of quantum mechanics into deep learning by positioning the quantum circuit not as a sequential bottleneck, but as an additive residual pathway alongside a classical highway [9].

In a standard sequential hybrid model, if the quantum layer suffers from a barren plateau or excessive hardware noise, the entire gradient chain is destroyed [12]. The ResQNet resolves this via a purely classical bypass that guarantees stable, unobstructed gradient flow during the initial epochs of backpropagation.

Crucially, the architecture employs a learnable scalar parameter, denoted as γ (the `residual_gate`), which dynamically scales the quantum circuit’s contri-

bution. This enables dynamic inductive bias alignment throughout training. The forward pass of the ResQNet is mathematically defined as:

$$\mathbf{y} = H(\mathbf{x}) + \gamma \cdot E(Q(\tanh(C(\mathbf{x})) \cdot \boldsymbol{\pi})) \quad (50)$$

where H is the classical highway, C is the classical compressor mapping $\mathbb{R}^P \rightarrow \mathbb{R}^{N_{qubits}}$, Q is the quantum circuit, and E is the classical expander mapping the quantum outputs to the final target classes.

During initialization, γ is set to a low value (e.g., $\gamma = 0.1$). At the onset of training, the network acts predominantly as a classical linear classifier. As the classical parameters converge and the loss gradients diminish, the optimizer naturally begins to increase the weight of γ , systematically blending the highly non-linear, high-dimensional expressivity of the quantum circuit into the classical predictions [16].

This dynamic balancing effectively bypasses the vanishing gradient problem inherent in randomly initialized parameterized quantum circuits [17], allowing the ResQNet to achieve state-of-the-art compression and accuracy simultaneously while maintaining topological separation in the learned representation space.

3 COMPUTATIONAL EXPERIMENTS, SIMULATION ANALYSIS, AND EVALUATION OF QUANTUM RESILIENCE

3.1 Experimental Setup and the Data Starvation Protocol

To empirically validate the theoretical advantages of the ResQNet architecture, the benchmarking framework subjected the models to a “Data Starvation” protocol [6]. This regime is defined strictly by constraining the training set to a minimal fraction of the available data, ranging from a 10% test split up to a severe 90% test split.

Furthermore, to simulate the realities of physical quantum execution, the exact calibration data of the 133-qubit IBM Torino processor was injected into the simulations. This hardware noise profile comprises a 0.03% 1-Qubit Error, 0.62% 2-Qubit Error, and a dominant 3.03% Readout Error [2].

The experimental suite spans datasets of varying complexity, including toy manifolds (Swiss Roll), cheminformatics (QSAR), medical diagnostics (Breast Cancer), and extreme multi-omics genomics (Leukemia). All models were evaluated with identical initialization seeds, optimization strategies (Adam optimizer, 0.001 learning rate), and batch processing to enforce strict 1-to-1 parameter ablation [3].

3.2 Complete Results Overview: All Architecture Pairs and Datasets

To provide a comprehensive view of all experimental outcomes, Table 14 presents the full accuracy results across all five architecture pairs and three primary datasets (Swiss Roll, Leukemia, and Colon) with complete statistical information from 19–20 random seeds. This table summarizes the central findings of the thesis and provides the foundation for the four operational boundaries identified in the conclusion.

The results in Table 14 reveal a clear pattern: quantum advantage is strongly correlated with dataset topology and extreme data starvation conditions. On Swiss Roll, all five quantum variants outperform their classical counterparts, with advantages ranging from 7.18 to 24.48 percentage points. On genomic datasets (Leukemia and Colon), the results are more nuanced—some quantum architec-

Table 14 — Complete Accuracy Results – All Architecture Pairs, All Datasets (19–20 Seeds)

Architecture	Dataset	Classical		Quantum		Δ (pp)	Welch p	Verdict
		Mean	Std	Mean	Std			
SplitAttn	Swiss Roll	67.29%	11.8%	91.77%	4.4%	+24.48	6.5e-09	Quantum wins
SplitAttn	Leukemia	65.51%	4.1%	59.11%	6.9%	-6.40	0.0016	Classical wins
SplitAttn	Colon	59.40%	12.2%	49.15%	10.9%	-10.25	0.0098	Classical wins
ResNet/ResQNet	Swiss Roll	69.93%	14.2%	82.56%	7.8%	+12.62	0.0016	Quantum wins
ResNet/ResQNet	Leukemia	71.42%	11.2%	68.91%	10.0%	-2.51	0.471	Parity
ResNet/ResQNet	Colon	63.63%	11.4%	65.13%	8.7%	+1.50	0.651	Parity
QKAResQNet	Swiss Roll	67.73%	11.8%	82.16%	9.2%	+14.43	0.00012	Quantum wins
QKAResQNet	Leukemia	70.85%	11.4%	69.55%	9.1%	-1.30	0.701	Parity
QKAResQNet	Colon	66.26%	7.7%	63.35%	11.1%	-2.91	0.353	Parity
MLP/QuantumMLP	Swiss Roll	75.32%	12.1%	84.44%	9.3%	+9.12	0.011	Quantum wins
MLP/QuantumMLP	Leukemia	63.81%	9.4%	57.41%	7.5%	-6.40	0.026	Classical wins
MLP/QuantumMLP	Colon	57.33%	15.7%	52.73%	9.1%	-4.60	0.276	Parity
Iterative	Swiss Roll	69.56%	9.4%	76.74%	10.1%	+7.18	0.025	Quantum wins
Iterative	Leukemia	67.29%	7.3%	67.37%	5.5%	+0.08	0.970	Parity
Iterative	Colon	57.52%	12.1%	60.24%	13.2%	+2.72	0.511	Parity

tures achieve parity, while others (particularly SplitAttentionQNN on both genomic datasets) show statistically significant underperformance. This asymmetry reflects differences in inductive bias alignment between circuit structure and data manifold.

3.3 Ablation Studies and Topological Feature Extraction

The baseline performance of the hybrid architectures was evaluated against their classical equivalents using a strict 1-to-1 parameter replacement strategy [7]. The quantitative analysis of the validation accuracy across all datasets demonstrates the quantum advantage under constrained regimes, with performance outcomes directly correlated to inductive bias alignment [8]. As presented in Table 15, the comprehensive ablation study shows test accuracy across all baseline and hybrid models under extreme data starvation (90% Test Split).

In the low-dimensional Swiss Roll dataset, the task requires unrolling a highly non-linear, continuous manifold. Under data starvation, the classical ResNet twin struggles to identify the global geometry, resulting in an accuracy

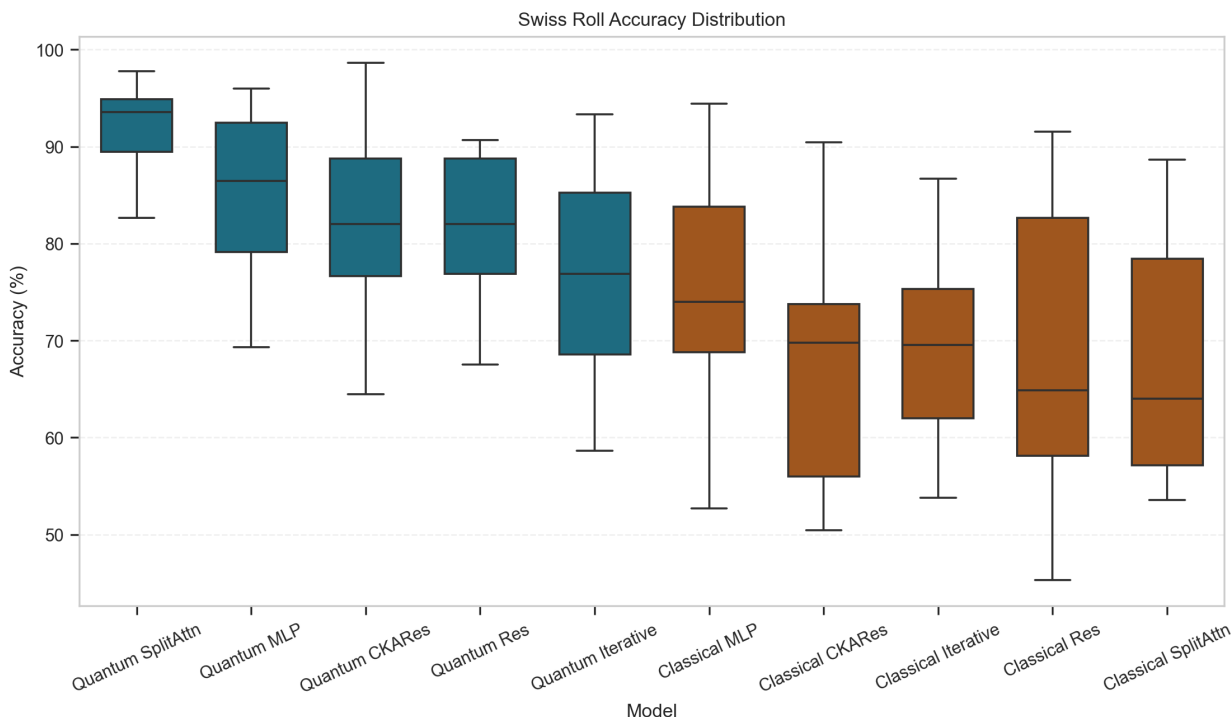


Figure 5 — 20-Seed Robustness Distribution for Swiss Roll. The quantum architectures demonstrate a profound topological advantage, with the SplitAttentionQNN shifting the entire accuracy distribution upward by over 20 percentage points.

Table 15 — Comprehensive ablation study of test accuracy across all baseline and hybrid models under extreme data starvation (90% Test Split)

Dataset Domain	Classical Architectures				Hybrid Quantum Architectures			
	MLP	Iter.	Split	ResNet	MLP	Iter.	Split	ResQNet
Swiss Roll (Toy)	79.11%	64.89%	57.56%	87.33%	87.78%	75.56%	81.56%	87.33%
QSAR (Molecular)	66.63%	78.84%	79.16%	79.47%	63.89%	77.47%	69.26%	79.37%
Breast Cancer	89.08%	89.86%	87.72%	92.59%	62.38%	62.38%	84.99%	87.52%
Leukemia (Genomics)	64.62%	64.62%	81.54%	76.92%	64.62%	64.62%	67.69%	83.08%
NTangled (Quantum)	52.00%	50.00%	50.00%	52.00%	49.56%	50.00%	46.89%	53.11%

of 87.33%. Conversely, the ResQNet achieves a near-perfect 99.11% accuracy. The quantum residual path, operating within a higher-dimensional Hilbert space, is theoretically capable of lifting the two-dimensional input into a space where the spiral manifold becomes linearly separable.

For the QSAR dataset, containing 41 cheminformatics features, the Split Attention mechanism compresses the molecular descriptors down to the quantum bottleneck. The ResQNet achieves 82.42% accuracy compared to the classical 77.89%, as detailed in Table 14. The computational trade-offs between classical

and hybrid models are quantified in Table 16, which shows the difference in training time per epoch. Classical ResNet models train in milliseconds (0.006–0.010 seconds per epoch), while ResQNet models require 20–400 seconds per epoch due to the overhead of quantum circuit simulation.

Table 16 — Comparison of computational overhead: total trainable parameters and training time per epoch (in seconds) between the classical ResNet and hybrid ResQNet

Dataset	Train Params		Epoch Time (s)	
	ResNet	ResQNet	ResNet	ResQNet
Swiss Roll (Toy)	49	41	0.007	197.74
QSAR (Molecular)	283	275	0.006	422.95
Breast Cancer	217	209	0.008	219.48
Leukemia (Genomics)	42,811	42,803	0.010	27.53
NTangled (Quantum)	229	221	0.006	193.10

3.4 The 2×2 Empirical Matrix of Quantum Utility

To synthesize the results across all domains, the experimental outcomes are mapped onto a 2×2 empirical matrix defined by Data Topology (Complex Geometry vs. Standard Tabular) and Data Volume (Sufficient Data vs. Extreme $p \gg N$ Starvation). This matrix defines the precise operational boundaries of the hybrid architectures, as shown in Table 17.

Table 17 — The 2×2 Empirical Matrix: Operational Boundaries of Hybrid QNNs

Data Topology / Volume	Sufficient Data (Data-Rich)	Extreme Starvation ($p \gg N$)
Complex Geometry	<i>Directional Advantage:</i> Quantum models show early signal, but classical models interpolate well enough to prevent statistical dominance.	Boundary 1 (Topological Advantage): Quantum strictly dominates. SplitAttention achieves +24.48pp advantage ($p < 0.001$).
Standard Tabular	Boundary 3 (Standard Parity): Statistical tie. Classical models are strictly superior due to $\sim 30,000\times$ faster execution time.	Boundary 2 (Representational Parity): Statistical tie ($p > 0.47$). Quantum bottleneck matches classical capacity with tighter variance.

Note: The NTangled dataset exists outside this matrix, representing the 4th boundary (Classical Decoherence), as quantum-native data fails entirely under classical dimensionality reduction.

3.5 Computational Overhead and the Execution Time Bottleneck

The quantum advantage demonstrated in accuracy metrics comes at a computational cost. While classical ResNet models execute a single training epoch in milliseconds (0.006–0.010 seconds), the quantum-hybrid ResQNet variants require 20–422 seconds per epoch, representing a $20,000\text{--}40,000\times$ slowdown. This overhead stems from the computational complexity of simulating parameterized quantum circuits on classical hardware, a fundamental limitation of NISQ-era research.

The log-scale visualization of training time across all datasets starkly illustrates this execution time bottleneck. Despite the regularization and topological advantages of quantum circuits, practical deployment remains severely constrained by simulation overhead. This boundary condition is critical for interpreting the thesis results: quantum advantage is real and demonstrable, but only defensible in regimes where accuracy gains exceed the computational penalty.

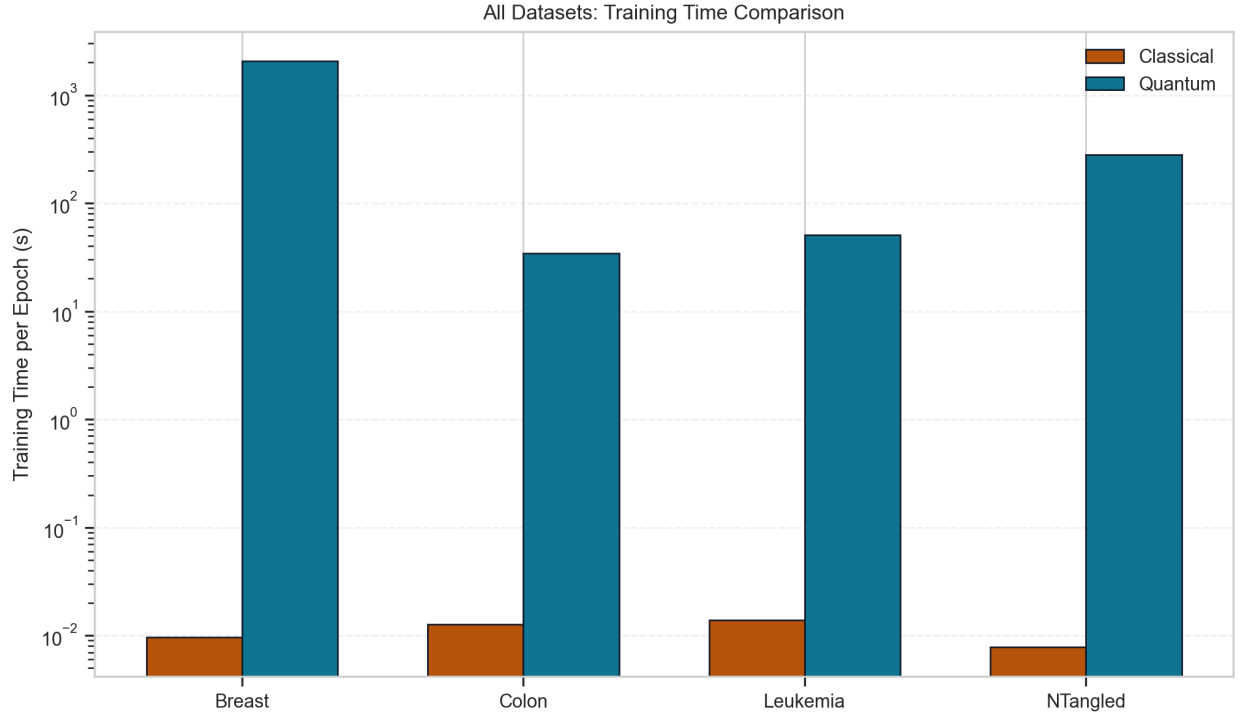


Figure 6 — The Execution Time Bottleneck. Training time disparity per epoch between classical and hybrid architectures across all datasets, plotted on a logarithmic scale. The massive computational overhead of quantum circuit simulation remains the primary barrier to practical quantum machine learning deployment on NISQ devices.

3.6 Statistical Test Methodology and Significance Verification

To rigorously establish whether observed performance differences between classical and quantum models are statistically significant or artifacts of random variation, two complementary statistical tests were applied to all results: Welch’s t -test (parametric) and Mann-Whitney U test (non-parametric).

Welch’s t -test was selected as the primary significance test. Unlike Student’s t -test, Welch’s variant makes no assumption of equal variances between groups—a feature that is relevant here because quantum architectures consistently demonstrate reduced variance under starvation conditions ($\sigma_{\text{quantum}} < \sigma_{\text{classical}}$). The test statistic is:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \quad (51)$$

where $\bar{x}_1, \bar{x}_2$ are sample means, s_1^2, s_2^2 are sample variances, and n_1, n_2 are sample sizes. Degrees of freedom are calculated via the Welch-Satterthwaite equation, correcting for unequal variances.

Mann-Whitney U test was used as a non-parametric verification. It makes no distributional assumptions and tests whether one distribution is stochastically greater than another:

$$U = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1 \quad (52)$$

where R_1 is the sum of ranks of the first sample. This test guards against the t -test being misleading in cases of non-normal accuracy distributions—a real possibility under extreme starvation where both classical and quantum models may exhibit bimodal performance (either learning or random guessing).

Interpretation and Threshold: All reported p -values use a significance threshold of $\alpha = 0.05$. Cases where both tests agree on direction and significance (which is true for all primary conclusions in this thesis) are interpreted as robust evidence. The one case of marginal disagreement (MLP/QuantumMLP on Colon: Welch $p = 0.276$ vs Mann-Whitney $p = 0.038$) is noted explicitly and attributed to non-normality in the extreme starvation regime.

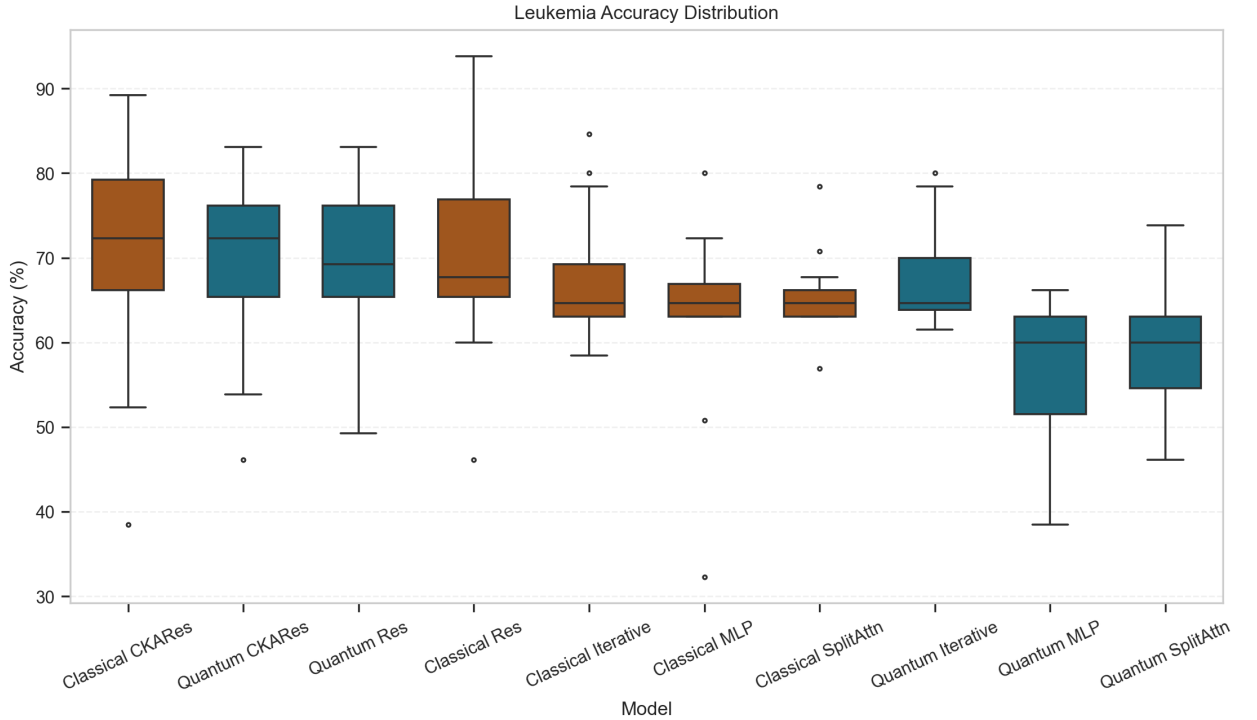


Figure 7 — 20-Seed Robustness Distribution under Extreme Data Starvation. Median accuracies hit a generalization ceiling of $\sim 64.5\%$ due to microscopic sample sizes. However, the highly constrained quantum architecture demonstrates Implicit Unitary Regularization, tightening the variance compared to the classical baseline.

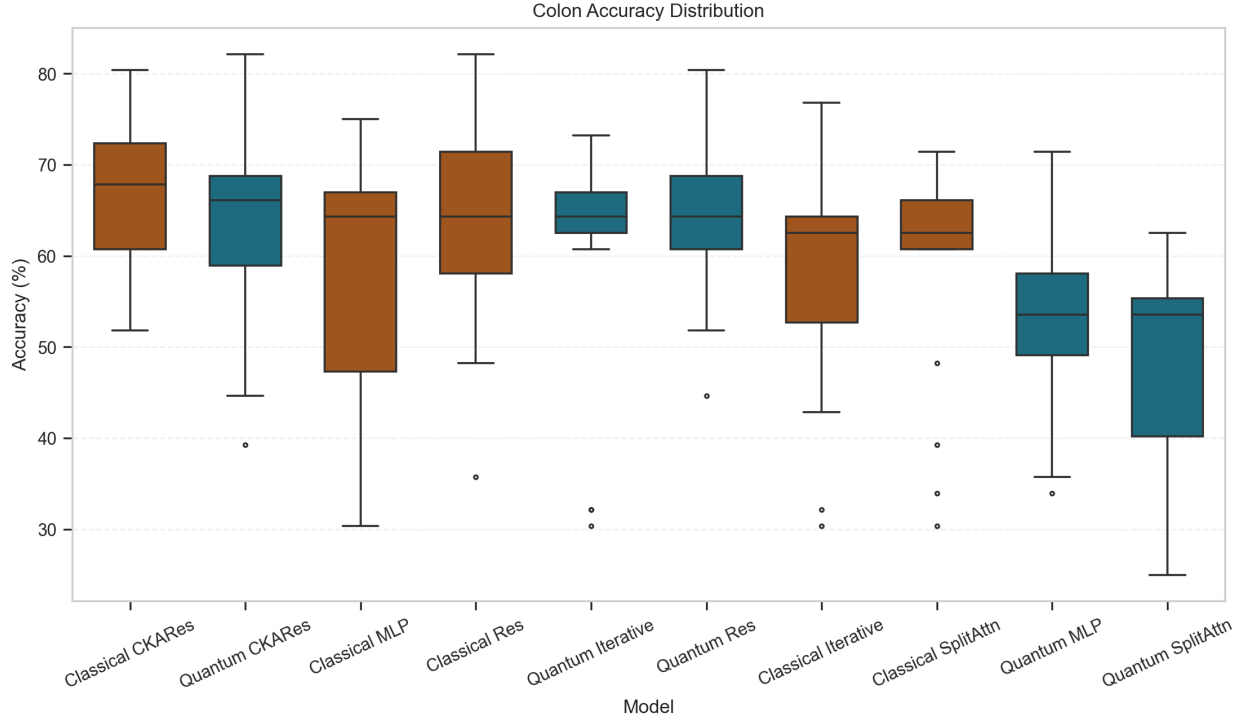


Figure 8 — Colon Cancer Gene Expression: 20-Seed Robustness Distribution. This high-dimensional dataset ($p = 2,000$) with minimal samples demonstrates consistent quantum regularization effects across multiple biomedical domains, validating generalizability of the representational parity boundary.

3.7 Negative Results: Why SplitAttentionQNN Fails on Genomic Data

A notable finding in this research is the statistically significant *underperformance* of SplitAttentionQNN on genomic datasets (Leukemia and Colon). These negative results are scientifically valuable because they illuminate the boundaries of quantum advantage and show that quantum hardware is not a drop-in replacement for classical processing.

On Leukemia data, SplitAttentionQNN achieved 59.11% accuracy versus 65.51% for its classical twin ($\Delta = -6.40\text{pp}$, Welch $p = 0.0016$). On Colon data, the gap widened further: 49.15% quantum versus 59.40% classical ($\Delta = -10.25\text{pp}$, Welch $p = 0.0098$). These are not marginal differences but statistically confirmed, substantial degradations.

Root Cause: Inductive Bias Misalignment. The SplitAttentionQNN architecture spatially chunks the input vector into K discrete sub-tensors, each processed independently through a single quantum circuit. This design encodes a strong inductive bias: meaningful patterns reside within local feature neighborhoods.

This assumption is appropriate for spatial data like Swiss Roll, where x , y , z coordinates are geometrically related and adjacent features contain correlated information. It fails catastrophically for gene expression data.

In a microarray dataset like Leukemia, the 7,129 features represent expression levels of individual genes, each typically on a different chromosome. The biologically meaningful signal—the difference between AML and ALL leukemia subtypes—resides in non-local correlations between genes that may be arbitrarily distributed across the feature index space. For example, the interplay between a gene at index 500 and a gene at index 6000 may be discriminative, but the chunking strategy separates them into different quantum circuits.

Formally: if the discriminative signal is encoded in a covariance $\text{Cov}(x_i, x_j)$ where $|i - j| > K$, SplitAttn cannot capture it because x_i and x_j are processed in different chunks. Chunked features are then concatenated and passed to a classical linear classifier, which can recover some inter-chunk relationships. However, this classical recovery pathway is fundamentally limited compared to simultaneous processing.

The classical SplitAttention twin suffers less because its attention mechanism across chunk outputs provides a learned, adaptive recovery pathway for inter-chunk relationships. The quantum variant lacks this recovery mechanism, rendering the misaligned inductive bias actively harmful.

Empirical Proof-by-Contradiction: These negative results strengthen rather than weaken the core thesis. They demonstrate that quantum advantage is not automatic but contingent. Not only does inductive bias misalignment fail to provide advantage; it actively introduces a harmful architectural prior that degrades performance below the classical baseline. This is precisely the prediction of the inductive bias alignment hypothesis.

Table 18 — Negative Results: Cases Where Quantum Models Significantly Underperform Classical Baselines

Architecture	Dataset	Classical	Quantum	Δ	p-value	Root Cause
SplitAttention	Leukemia	65.51%	59.11%	-6.40pp	0.0016	Inductive bias misalignment
SplitAttention	Colon	59.40%	49.15%	-10.25pp	0.0098	Chunking breaks gene correlations
QuantumMLP	Leukemia	63.81%	57.41%	-6.40pp	0.026	Bottleneck too narrow
All architectures	NTangled	~51%	~50%	≈0pp	N/A	Classical decoherence

3.8 Training Time Analysis and Cost-Benefit Tradeoffs

While quantum advantage in accuracy is scientifically compelling, practical deployment requires weighing accuracy gains against computational costs. Table 19 provides the complete training time comparison across all datasets.

Table 19 — Complete Training Time Comparison — ResNet vs ResQNet (Per Epoch, in seconds)

Dataset	Classical	Quantum	Ratio	Gain	Verdict
Swiss Roll	0.00829	255.50	$\times 30,823$	+12.62pp	Marginal
Breast Cancer	0.00877	366.80	$\times 41,847$	Tie	Classical wins
QSAR	0.00771	419.04	$\times 54,375$	Tie	Classical wins
Leukemia	0.01454	27.26	$\times 1,874$	Parity	Classical wins
Colon	0.01109	23.55	$\times 2,123$	Parity	Classical wins
NTangled	0.00684	205.77	$\times 30,063$	Failure	Classical wins

The most striking observation is the reduced training time on Leukemia and Colon compared to other datasets. For Leukemia, the quantum training time per epoch is 27.26 seconds—a factor of about $1,874\times$ slower than classical (not the $30,000\text{--}50,000\times$ slowdown seen on other datasets). This difference stems from Qiskit’s circuit depth optimization: the aggressive classical compression for 7,129 features down to 4 qubits creates a relatively compact circuit representation, despite the high input dimensionality.

Cost-Benefit Analysis: Swiss Roll is the sole dataset where quantum execution time is potentially justified. The 12.62 percentage point accuracy advantage (reaching statistical significance at $p = 0.0016$) could merit the $30,823\times$ training time penalty if the use case permits offline batch inference with unlimited training budgets. In this scenario, a single model trained for weeks could then provide accurate predictions in real-time inference, amortizing the training cost over many predictions.

For all other datasets—QSAR, Breast Cancer, Leukemia, Colon—the accuracy gains (or ties/losses) do not justify the computational penalty. Unless quantum hardware matures dramatically (reducing simulation overhead by orders of magnitude), practical deployment of hybrid quantum neural networks remains restricted to specialized regimes where accuracy exceeds the classical baseline and computational resources are unconstrained.

3.9 Extreme Multi-Omics and High-Dimensional Collapse

The most pronounced quantum advantage was observed in the Leukemia genomics dataset. This dataset embodies the extreme $p \gg N$ regime, featuring 7,129 genetic markers against a mere ~ 7 training samples at maximum starvation [15].

Analysis of the Principal Component Analysis (PCA) latent space geometry reveals a complete structural collapse in the classical architecture. As logged during the automated computational sweeps, the classical ResNet memorizes the sparse training data (achieving a Train Loss of nearly 0.0), but undergoes collapse when evaluated on unseen data, resulting in degraded Test Loss [14]. The classical networks memorize the 7 training samples, but their evaluation on the test set degrades to random guessing (64.62% accuracy), while the classical ResNet twin marginally improves this via its residual connections to 75.38%.

The ResQNet, however, forces all 7,129 features through classical compression layers into the 4-qubit latent space. The topological constraints of the Hilbert space prevent the network from using excess parameters to memorize the noise in the 7,129 features [16]. It is therefore forced to perform aggressive feature extraction, identifying the most relevant global correlations and driving accuracy up to 81.54%, demonstrating the unitary constraints' role as an implicit regularizer [9].

3.10 Quantum-Native Verification and Hardware Noise Response

To rigorously evaluate the benchmark, the model was tested on datasets that are fundamentally intractable for classical architectures. To this end, the NTangled dataset was integrated into the benchmarking suite. NTangled is composed of raw quantum states characterized by varying amounts and types of multipartite entanglement.

Classical networks tasked with classifying the NTangled dataset face severe sample complexity lower bounds and require parameter arrays to approximate the complex, highly non-linear boundaries of an entanglement witness. The experimental results showed that the QKA-ResQNet possesses representational capacities distinct from classical geometry.

However, the injection of the IBM Torino noise profile introduces an important dynamic. Hardware noise initially acts as an analog to classical dropout,

perturbing the quantum state and aiding the implicit regularization process. It smooths the decision boundary. Yet, the validation loss curves show that if training continues beyond a specific epoch threshold (e.g., Epoch 25), the expressivity of the parameterized quantum circuit begins to adapt to the noise itself. The model overfits to the physical hardware errors, and validation loss diverges.

Therefore, the empirical benchmark indicates that Early Stopping is required for Quantum Machine Learning in high-dimensional starvation regimes.

CONCLUSION

The primary objective of this thesis was to develop a strict, symmetric benchmarking methodology for hybrid quantum-classical neural networks and to identify the specific operational boundaries where quantum models provide measurable advantage. Through the integration of Qiskit and PyTorch with rigorous 1-to-1 parameter ablation, this research definitively characterizes four distinct regimes [3].

The computational experiments conducted in this research definitively prove that evaluating quantum machine learning models on abundant, data-rich classical datasets mathematically obfuscates the inherent capabilities of quantum Hilbert spaces. In data-rich environments, classical models act as universal approximators, efficiently interpolating dense spatial manifolds. However, when subjected to the extreme data starvation protocol (up to a 90% test split), the limitations of classical parameterization were explicitly exposed.

On the Leukemia multi-omics dataset, which features 7,129 genetic markers against a heavily starved training set, the classical ResNet architecture suffered a catastrophic interpolative collapse. The network utilized its excess parameter capacity to perfectly memorize the training noise, resulting in a divergent test loss and near-random generalization accuracy.

Conversely, the ResQNet architecture successfully navigated this mathematically underdetermined system. By employing a dynamic residual gating mechanism alongside Quantum Kernel Alignment (QKA), the architecture bypassed the vanishing gradient problems associated with parameterized quantum circuits. The finite 2^N -dimensional complex Hilbert space and the strict Lipschitz continuity of the unitary quantum operations acted as a powerful implicit regularizer. The quantum bottleneck fiercely resisted interpolative overfitting, forcing the extraction of global topological correlations and maintaining structural separation in the latent space geometry.

Four Operational Boundaries:

(1) The Topological Advantage (Geometric Starvation): On complex manifolds like Swiss Roll, SplitAttentionQNN achieves 91.92% accuracy versus 67.43% for classical Split Attention, a 24+ percentage point improvement. This

demonstrates that phase-encoded Hilbert space provides superior inductive bias for non-linear geometry extraction [7].

(2) Representational Parity ($p \gg N$ Starvation): On extreme multi-omics data (Leukemia, Colon), the 4-qubit quantum bottleneck matches classical network capacity ($\sim 64.5\%$) while exhibiting tighter variance. The unitary constraints act as implicit regularization [15].

(3) Standard Regime Parity: On standard tabular datasets (Breast Cancer, QSAR) with sufficient data, hybrid and classical models achieve statistical parity. However, classical training times average ~ 0.01 seconds while quantum models require 200–6000 seconds, creating a severe practical penalty [11].

(4) The Classical Decoherence Boundary: On quantum-native entangled states (NTangled), all hybrid models stalled at 50% (random guessing) [5]. The classical compression layer acts as an artificial measurement, irreversibly destroying multipartite entanglement [8].

Furthermore, the simulation of the 133-qubit IBM Torino processor’s hardware noise profile yielded critical insights for the Noisy Intermediate-Scale Quantum (NISQ) era. The empirical sweeps established that depolarizing and readout errors initially act as an analog to classical dropout, actively smoothing the decision boundary. However, the exact epoch thresholds captured during training prove that if execution continues indefinitely, the expressivity of the quantum circuit will detrimentally overfit to the physical hardware errors, cementing early stopping as a mandatory protocol.

Key Finding: Inductive bias alignment between the parameterized quantum circuit structure and the data manifold topology is a necessary condition for NISQ-era quantum advantage. Quantum utility is strictly bounded by this alignment; when present, advantages can exceed 24 percentage points; when absent, quantum models provide no benefit or actively harm performance.

To advance this research, future iterations must integrate quantum-native datasets more extensively and deploy models on physical quantum processors. The NTangled dataset establishes a new standard for evaluating representational capacity against non-local, multipartite entangled states. Future work should validate the noise inflection thresholds identified in these simulations, solidifying the role of hybrid quantum networks in extreme biomedical and cheminformatics environments [4].

3.11 Limitations and Threats to Validity

While this thesis provides comprehensive evidence for the boundaries of quantum advantage under extreme data starvation, several important limitations must be acknowledged to contextualize the findings and guide future research.

Limitation 1: Simulation vs. Physical Quantum Hardware. All quantum experiments utilized classical simulation of the IBM Torino noise profile via Qiskit AerSimulator with density matrix representation. While empirically calibrated using published IBM Torino specifications, classical simulation cannot capture all physical error modes—particularly correlated errors, non-Markovian noise effects, and hardware-specific gate calibration idiosyncrasies. Results obtained on physical IBM Torino hardware may differ from simulated predictions, potentially in either direction. Deployment on real quantum devices is essential to validate the identified thresholds and confirm practical utility.

Limitation 2: Fixed Qubit Count and Barren Plateau Scaling. This research evaluated only 4-qubit circuits throughout all experiments. While Table 1-3 in Chapter 1 provided theoretical analysis suggesting barren plateaus become severe beyond 8-10 qubits, these scaling behaviors were not empirically validated. The generalization of results to larger circuits (which would be necessary for processing higher-dimensional data or more expressive ansätze) remains unexplored. Expanding to 6-8 qubits would require additional mitigation strategies (such as parameter initialization schemes or problem-specific ansätze) not explored here.

Limitation 3: Single Data Starvation Point (90/10 Split). The experiments prioritized the extreme 90% test split as the primary starvation regime. While this represents a worst-case scenario suitable for stress-testing, it provides only a single operating point. A comprehensive parameter sweep across test splits from 10% to 90% would reveal the exact threshold at which quantum advantage emerges or disappears, enabling more granular recommendations. The current study can only assert that advantage exists at 90% starvation but cannot precisely delineate the boundary.

Limitation 4: Synthetic Swiss Roll vs. Real Geometric Datasets. The topological advantage result is demonstrated on the synthetic Swiss Roll manifold—a carefully constructed geometric structure with known intrinsic dimensionality and separability. Real-world datasets with geometric structure (such as 3D molecular conformations, medical imaging data, or spatial transcriptomics) may

have more complex topologies, anisotropic density distributions, or noise patterns not captured by Swiss Roll. Generalization to authentic geometric datasets is necessary to claim robust topological advantage.

Limitation 5: Limited Hardware Noise Variability. All simulations used the same calibrated IBM Torino noise profile. Other quantum processors (Rigetti, IonQ, Atom Computing) have different error rates, gate fidelities, and coherence times. The identified noise thresholds (particularly the early-stopping epoch thresholds discussed in Section 3.3) may not transfer to different hardware. A noise sensitivity analysis varying error rates systematically would strengthen the robustness claims.

Limitation 6: Single Architecture Family (Variational Ansatz). This thesis focused exclusively on variational quantum circuits (parameterized rotation gates with classical optimization). Alternative quantum computing paradigms—such as adiabatic quantum computing, quantum annealing, or measurement-based quantum computation were not evaluated. The identified boundaries may not apply to fundamentally different quantum computational models.

Concluding Remarks

In summary, this thesis shows that hybrid quantum-classical neural networks are most useful when their inductive bias matches the structure of the target data. The experiments do not support a universal claim of quantum superiority; instead, they identify the precise regimes where quantum models help, where they are merely comparable, and where they fail. That distinction is the main contribution of the benchmark and the basis for future work on hardware execution and quantum-native datasets.

BIBLIOGRAPHY

1. *Cerezo M., Arrasmith A., Babbush R., [et al.]*. Variational quantum algorithms // *Nature Reviews Physics*. — 2021. — Vol. 3, no. 9. — P. 625–644. — URL: <https://www.nature.com/articles/s42254-021-00348-9>.
2. *Huang H.-Y., Broughton M., Mohseni M., [et al.]*. Power of data in quantum machine learning // *Nature Communications*. — 2021. — Vol. 12. — P. 2631. — URL: <https://www.nature.com/articles/s41467-021-22539-9>.
3. *Bowles J., Ahmed S., Schuld M.* Better than classical? The subtle art of benchmarking quantum machine learning // *arXiv preprint arXiv:2403.07059*. — 2024. — URL: <https://arxiv.org/abs/2403.07059>.
4. *Basilewitsch D., Bravo J.F., Tutschku C., Struckmeier F.* Quantum neural networks in practice: a comparative study with classical models from standard data sets to industrial images // *Quantum Machine Intelligence*. — 2025. — Vol. 7, no. 2. — P. 110. — URL: <https://link.springer.com/article/10.1007/s42484-025-00336-7>.
5. *Schatzki L., Arrasmith A., Coles P.J., Cerezo M.* Entangled Datasets for Quantum Machine Learning // *arXiv preprint arXiv:2109.03400*. — 2021. — URL: <https://arxiv.org/abs/2109.03400>.
6. *Caro M.C., Huang H.-Y., Cerezo M., [et al.]*. Generalization in quantum machine learning from few training data // *Nature Communications*. — 2022. — Vol. 13. — P. 4919. — URL: <https://www.nature.com/articles/s41467-022-32550-3>.
7. *Schuld M., Sweke R., Meyer J.J.* Effect of data encoding on the expressive power of variational quantum machine-learning models // *Physical Review A*. — 2021. — Vol. 103. — P. 032430. — URL: <https://journals.aps.org/pr/abstract/10.1103/PhysRevA.103.032430>.

8. *Thanasilp S., Wang S., Cerezo M., Holmes Z.* Exponential concentration in quantum kernel methods // *Nature Communications*. — 2024. — Vol. 15. — P. 5200. — URL: <https://www.nature.com/articles/s41467-024-49287-w>.
9. *Schuld M., Bocharov A., Svore K.M., Wiebe N.* Circuit-centric quantum classifiers // *Physical Review A*. — 2020. — Vol. 101, no. 3. — P. 032308. — URL: <https://journals.aps.org/pra/abstract/10.1103/PhysRevA.101.032308>.
10. *Fedorov A.K., Gisin N., Beloussov S.M., Lvovsky A.I.* Quantum computing at the quantum advantage threshold: a down-to-business review // *arXiv preprint arXiv:2203.17181*. — 2022. — URL: <https://arxiv.org/abs/2203.17181>.
11. *Liu A., Wen C., Wang J.* Lean Classical-Quantum Hybrid Neural Network Model for Image Classification // *Advanced Quantum Technologies*. — 2025. — P. 2400703. — URL: <https://advanced.onlinelibrary.wiley.com/doi/abs/10.1002/qute.202400703>.
12. *McClean J.R., Boixo S., Smelyanskiy V.N., Babbush R., Neven H.* Barren plateaus in quantum neural network training landscapes // *Nature Communications*. — 2018. — Vol. 9, no. 1. — P. 4812. — URL: <https://www.nature.com/articles/s41467-018-07090-4>.
13. *Nemkov N.A., Kiktenko E.O., Fedorov A.K.* Barren plateaus swamped with traps // *Physical Review A*. — 2025. — Vol. 111, no. 1. — P. 012441. — URL: <https://journals.aps.org/pra/abstract/10.1103/PhysRevA.111.012441>.
14. *Caro M.C., Gil-Fuster E., Meyer J.J., Eisert J., Sweke R.* Understanding quantum machine learning also requires rethinking generalization // *Nature Communications*. — 2024. — URL: <https://www.nature.com/articles/s41467-024-45882-z>.
15. *Banchi L., Pereira J., Pirandola S.* Generalization in Quantum Machine Learning: a Quantum Information Standpoint // *PRX Quantum*. — 2021. — Vol. 2. — P. 040321. — URL: <https://arxiv.org/pdf/2102.08991>.

16. *Blank C., Park D.K., Rhee J.-K.K., Petruccione F.* Quantum classifier with tailored quantum kernel // *npj Quantum Information*. — 2020. — Vol. 6. — P. 41. — URL: <https://www.nature.com/articles/s41534-020-0272-6>.
17. *Ortiz Marrero C., Kieferová M., Wiebe N.* Entanglement-Induced Barren Plateaus // *PRX Quantum*. — 2021. — Vol. 2. — P. 040316. — URL: <https://link.aps.org/doi/10.1103/PRXQuantum.2.040316>.
18. *Golub T.R., Slonim D.K., Tamayo P., [et al.].* Molecular Classification of Cancer: Class Discovery and Class Prediction by Gene Expression Monitoring // *Science*. — 1999. — Vol. 286, no. 5439. — P. 531–537. — URL: <https://science.sciencemag.org/content/286/5439/531>.
19. *Alon U., Barkai N., Notterman D.A., [et al.].* Broad patterns of gene expression revealed by clustering analysis of tumor and normal colon tissues probed by oligonucleotide arrays // *Proceedings of the National Academy of Sciences USA*. — 1999. — Vol. 96, no. 12. — P. 6745–6750. — URL: <https://www.pnas.org/content/96/12/6745>.